

The Welfare of Nations: Revealed Social Preferences in Incomplete-Market Economies*

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We estimate social preferences from observed fiscal systems, developing an inverse-optimal welfare approach for heterogeneous-agent economies with incomplete markets. In a stationary Bewley economy, standard Pareto-Negishi social welfare functions converge to the utilitarian benchmark because long-run heterogeneity is generated by idiosyncratic histories. We therefore introduce a new history-based class of social welfare functions, with both an axiomatic characterization and an aggregation interpretation. Applying the framework to fiscal systems in France and the United States, we recover social welfare weights over current productivity states. The recovered valuations differ sharply across countries: US weights rise with productivity, while French weights are more compressed and U-shaped, with the highest valuations assigned to low-productivity states and a smaller secondary rise at the top. These patterns are robust to alternative utility specifications, non-parametric recovery, and history-dependent extensions.

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1 Introduction

Heterogeneous-agent models with incomplete markets have become a standard framework for modern macroeconomic analysis. By generating endogenous heterogeneity through uninsurable idiosyncratic risk, these models provide a realistic description of the distributions of income, wealth, and consumption observed in the data. As these models become increasingly popular tools for normative policy evaluation, they also force a choice about how gains and losses across heterogeneous households should be aggregated together.

This raises a simple but consequential question: which social welfare function (SWF) should the planner use? The utilitarian benchmark assigns the same social weight to each household's utility. This benchmark is transparent and widely used, but it also fixes one particular normative criterion before looking at the data. In a frictionless benchmark, this criterion would lead the planner to equalize marginal utilities across households. More generally, it gives equal utility weights independently of the histories that generate heterogeneity in incomplete-market economies. We therefore depart from this benchmark by introducing richer welfare criteria and measuring how far the data move us away from utilitarianism.

From a practical standpoint, we compute the welfare weights implied by observed fiscal systems. To do so, we solve the Ramsey problem in a heterogeneous-agent model and recover the welfare criterion that rationalizes observed inequality and fiscal policy. This inverse-optimal approach has been widely used in static environments. We extend it to a dynamic Ramsey setting in general equilibrium. The recovered weights should be read as a revealed planner criterion, conditional on the model, the instruments under consideration, and the planner's commitment. In this sense, the exercise may also reflect political and institutional frictions that shape observed policy. We view the approach as a disciplined way to extract revealed social valuations from observed allocations.

This paper makes two contributions. The first contribution is theoretical. We develop a class of social welfare functions that is well suited to heterogeneous-agent models. The formal analysis provides an axiomatic characterization of this history-based criterion in a simple two-period environment and then extends the construction to an infinite-horizon heterogeneous-agent economy. The standard way of making utilitarianism more flexible is insufficient in stationary incomplete-market economies. A natural non-utilitarian approach assigns fixed Pareto-Negishi weights to agents according to their identities or initial characteristics.¹ The difficulty is that long-run heterogeneity in these economies is generated by realized idiosyncratic shocks rather than by initial characteristics. We show that any social welfare function based only on initial characteristics converges to the utilitarian criterion in the long run. Initial conditions cease to play a role in determining

¹The utilitarian social welfare function is a special case, which assigns identical weights to all agents (Aiyagari, 1995 and Aiyagari and McGrattan, 1998).

stationary wealth and consumption outcomes, and therefore lose their relevance for social aggregation. Consequently, fixed Pareto-Negishi welfare criteria become observationally equivalent to utilitarianism in stationary equilibrium.

For this reason, our new class of social welfare functions assigns welfare weights to the realizations of idiosyncratic histories rather than to agents' initial identities. The key insight is that idiosyncratic histories are the primitive source of heterogeneity in incomplete-markets environments. A welfare criterion defined over histories therefore remains meaningful even when initial conditions are forgotten. For example, a low-productivity household today may be in that state because of a particular sequence of shocks, not because of its initial type. We first introduce a general history-based social welfare function, which allows welfare weights to depend on the entire realization of idiosyncratic shocks. We then characterize tractable restrictions in which welfare weights depend only on a finite history, and finally on the current idiosyncratic state. This last specification, which we call the G-SWF (for Generalized SWF), is the stationary welfare object that we take to the data. It provides a parsimonious and operational generalization of the utilitarian benchmark that can be readily implemented in quantitative models.

Our SWF construction also has a simple aggregation interpretation. Individuals may disagree about how society should evaluate allocations, and these views may themselves depend on personal economic experiences. Since economic experiences are summarized by the realization of idiosyncratic shocks, aggregating heterogeneous social valuations naturally generates welfare weights attached to histories. This gives a natural analogy with Bewley economies themselves: just as economic heterogeneity emerges from realized idiosyncratic shocks, heterogeneity in social evaluations can emerge from differences in experienced histories. Thus, the same histories that generate economic heterogeneity also provide a natural basis for social aggregation.

The second contribution is quantitative. We quantify the extent to which the welfare weights revealed by fiscal systems differ across countries. Using data for France and the United States, we recover the G-SWF weights that rationalize observed progressive labor taxation, capital taxation, consumption taxation, and public debt in each country. In the United States, recovered welfare weights increase with productivity, whereas in France they are more compressed and U-shaped, with the highest valuations assigned to low-productivity states and a smaller secondary rise at the top.

We also use the recovered weights to assess their quantitative implications for fiscal policy and inequality. In a counterfactual exercise, we simulate the US economy under the French G-SWF. Wealth inequality declines significantly, with the Gini coefficient falling from 78% to 63%, below the French level. This reflects the interaction between French social weights and US economic primitives. This exercise shows how differences in recovered welfare weights can translate into different fiscal systems and equilibrium inequality.

Because recovered G-SWF weights depend on the utility specification and the class of welfare criteria under consideration, we assess robustness along three dimensions: (i) we vary the utility specification; (ii) we replace the parametric weight profile with a non-parametric recovery; and (iii) we allow weights to depend on truncated histories rather than only on current productivity. These exercises show that the recovered profiles are qualitatively robust to the main specification choices behind the inverse-optimal approach.

Overall, the paper demonstrates that richer social welfare criteria can be operationalized in quantitative heterogeneous-agent models. By moving beyond welfare weights attached to identities and focusing instead on the histories that generate economic heterogeneity, the framework manages to recover revealed social valuations while preserving the quantitative discipline of modern heterogeneous-agent macroeconomics.

The paper proceeds as follows. The next subsection discusses the related literature. Section 2 develops the foundation of our SWF construction in a two-period environment. Section 3 extends the construction to an infinite-horizon Bewley economy. Section 4 embeds the G-SWF in the Ramsey problem and derives the inverse-optimal restrictions. Section 5 presents the France-US recovery. Finally, Section 6 discusses maintained assumptions and robustness, while Section 7 concludes.

Literature

This paper contributes to three connected strands of the literature: normative policy analysis in heterogeneous-agent economies, non-utilitarian social welfare criteria, and the empirical recovery of social welfare weights from observed policies.

The paper is related to the growing literature on normative analysis in heterogeneous-agent economies. Since the pioneering contributions of Aiyagari (1995) and Aiyagari and McGrattan (1998), heterogeneous-agent incomplete-market models have become a workhorse framework for studying optimal taxation, social insurance, public debt, and monetary policy. Examples include Conesa and Krueger (2006); Conesa et al. (2009); Heathcote (2005); Bhandari et al. (2017, 2021b); Dávila et al. (2012); Dyrda and Pedroni (2022); McKay and Wolf (2025); Auclert et al. (2024). Recent contributions propose decompositions and welfare assessment for a given SWF (Floden, 2001; Dyrda and Pedroni, 2022; Bhandari et al., 2021a; Dávila and Schaab, 2025; Olivi et al., 2025). These papers typically evaluate policy using an ex-ante perspective or Pareto-Negishi weights. We depart from this approach by emphasizing an implication of stationarity in Bewley economies: fixed Pareto-Negishi weights lose their non-utilitarian content in the long run and converge to the utilitarian benchmark. This motivates our use of a non-utilitarian welfare criterion in the inverse-optimal exercise.

A second strand of the literature studies non-utilitarian welfare criteria (Benabou, 2002; Saez and Stantcheva, 2016; Chang et al., 2018; McKay and Wolf, 2025). Our

welfare criterion attaches welfare weights to realized histories. The G-SWF restricts these weights to current productivity states for the quantitative exercise. The history-based interpretation is related to empirical work showing that personal experiences shape economic beliefs and preferences (Malmendier and Nagel, 2011, for instance).

Finally, this paper contributes to inverse-optimal welfare analysis, which recovers social welfare weights from observed tax and transfer systems. A growing body of work recovers social welfare weights from observed tax and transfer systems. Saez and Stantcheva (2016) develop generalized social marginal welfare weights as a positive and normative tool for tax analysis. Lockwood and Weinzierl (2016), Finkelstein and Hendren (2020), and Hendren (2020) infer welfare weights from policy choices and empirical sufficient statistics. Heathcote and Tsujiyama (2021) recover social welfare weights consistent with observed allocations in a static environment. We extend this inverse-optimal perspective to a dynamic general-equilibrium heterogeneous-agent model with taxation and public debt.

2 A history-based welfare foundation

We first introduce the three welfare objects in the simplest two-period environment: an unrestricted history-based SWF, a fixed-lifetime PN-SWF, and the G-SWF that will be taken to the data.

2.1 Environment

We consider a two-period economy with a finite set of observable productivity levels $y \in \mathcal{Y}$, populated by a mass $\pi_y > 0$ of agents at productivity level y , with $\sum_{y \in \mathcal{Y}} \pi_y = 1$. In period 1, a mass π_y of ex-ante identical agents starts at productivity level y and receives endowment $e_y > 0$. Agents with productivity level y in period 1 transition to productivity level y' in period 2 with probability $\Pi_{yy'} > 0$, where the transition matrix Π is irreducible and satisfies $\pi_{y'} = \sum_y \pi_y \Pi_{yy'}$ for each destination productivity level y' .

The period-1 consumption of an agent with productivity y is denoted $c_{1,y}$. The period-2 consumption of an agent with productivity y at period 1 and y' at period 2 is denoted $c_{2,y,y'}$. Indeed, in the general case, consumption is indexed by the whole realization of the idiosyncratic history. Due to the law of large numbers, the mass of agents with history y, y' at period 2 is $\pi_{y,y'} = \pi_y \Pi_{y,y'}$. The pair (y, y') is the simplest history: y is the initial productivity state and y' is the contemporaneous productivity state.

To focus on the simplest case, we assume that the planner has a complete set of non-distortionary tools. It can thus choose $c_{1,y}$ and $c_{2,y,y'}$ for all $y, y' \in \mathcal{Y}$. Total resources

in each period are $Y := \sum_{y \in \mathcal{Y}} \pi_y e_y$, so feasibility requires:

$$\sum_{y \in \mathcal{Y}} \pi_y c_{1,y} \leq Y \text{ and } \sum_{y,y' \in \mathcal{Y}} \pi_{y,y'} c_{2,y,y'} \leq Y, \quad (1)$$

Period utility is $u(c)$ with $u' > 0$, $u'' < 0$, and the discount factor is β .

2.2 The history-based SWF and its restrictions

We now introduce the unrestricted history-based social welfare function. In the two-period environment, the planner can assign one weight to each period-1 productivity state and one weight to each period-2 history. Let $\omega_{1,y}$ denote the weight attached to agents with productivity y in period 1, and let $\omega_{2,y,y'}$ denote the weight attached to agents with history (y, y') in period 2. The history-based SWF, or H-SWF, is:

$$\mathcal{W}^H = \sum_{y \in \mathcal{Y}} \pi_y \omega_{1,y} u(c_{1,y}) + \beta \sum_{y,y' \in \mathcal{Y}} \omega_{2,y,y'} \pi_{yy'} u(c_{2,y,y'}). \quad (2)$$

This object is unrestricted in the sense that period-2 welfare weights can depend on the full two-period history.

In Section 2.2.1, we first provide an axiomatization of the H-SWF. In Section 2.2.2, we interpret its weights through a microfoundation based on heterogeneous individual social evaluations. Because the H-SWF is very general, we next study two empirically relevant restrictions of the H-SWF. The first, detailed in Section 2.2.3, is the standard Pareto-Negishi weights, denoted PN-SWF, which collapses histories to initial productivity and assigns fixed lifetime weights to any agent. The second, detailed in Section 2.2.4, is the G-SWF, which collapses histories to contemporaneous productivity and assigns weights to current individual states.

2.2.1 Axiomatization of the H-SWF

The H-SWF applies familiar additive social-welfare logic to a different source of heterogeneity. In a usual heterogeneous-agent SWF, the planner may attach different social weights to different individuals or types; here the ethically relevant type is a history. This is close to the informational-basis view of social welfare, in which the social criterion first specifies which individual information is relevant and then aggregates it on a common welfare scale (Harsanyi, 1955; d'Aspremont and Gevers, 1977).

We take as given the common cardinal utility index u used to evaluate consumption in the model. The representation result takes this utility scale as given and asks what social criterion follows once allocations are evaluated through this common scale and the planner's heterogeneity source is histories. In the two-period environment, a history realization is either a period-1 productivity state y or a period-2 history (y, y') . The allocation therefore induces a finite vector of history-realization utilities, $u(c_{1,y})$ and

$u(c_{2,y,y'})$. The representation result rests on the following five conditions. First, the planner ranks allocations through the vector of history-realization utilities. Second, the criterion varies continuously with each history's utility. Third, it is monotone: raising utility in any history cannot reduce social welfare. Fourth, social trade-offs between histories do not depend on utility levels in other histories. Finally, we also impose that once the common cardinal utility unit is fixed, the planner treats a one-unit increase in a history-realization utility as having a constant social value. This condition makes the additive components affine in the history-realization utilities. These conditions are standard in the multi-attribute utility theory of Keeney and Raiffa (1976). The novelty is that they are applied here to histories (i.e., ex-post heterogeneity) rather than to types (i.e., ex-ante heterogeneity).

We can now state our representation result.

Proposition 1 (History-realization representation). *Under the conditions above, the planner criterion admits a history-realization representation: there exist nonnegative weights $(\omega_{1,y})_{y \in \mathcal{Y}}$ and $(\omega_{2,y,y'})_{y,y' \in \mathcal{Y}}$ such that, up to a positive normalization,*

$$\mathcal{W}^H = \sum_{y \in \mathcal{Y}} \pi_y \omega_{1,y} u(c_{1,y}) + \beta \sum_{y,y' \in \mathcal{Y}} \omega_{2,y,y'} \pi_{yy'} u(c_{2,y,y'}).$$

The proof is provided in Online Appendix A.1.

The proposition interprets history-based weights as the usual heterogeneous-agent SWF logic applied to histories rather than fixed individuals or contemporaneous states. Monotonicity gives nonnegative weights, while additive separability and the constant-social-value condition give the linear weighted form.

Like utilitarian and weighted-utilitarian criteria, this additive H-SWF is cardinal. It is invariant to common positive affine normalizations of the period utility or equivalently, to the induced common positive affine normalization of intertemporal utility. Its invariance is therefore limited to these common affine normalizations: arbitrary increasing transformations of period $u(\cdot)$ or intertemporal utility generally change the maintained welfare scale, as they do for utilitarian and weighted-utilitarian criteria. The representation result therefore makes explicit the cardinality maintained by additive social-welfare analysis.

2.2.2 A microfoundation through individual social evaluations

The axiomatization above is sufficient for the reduced-form H-SWF used in the paper. The following aggregation construction supplies one interpretation of where the H-SWF weights may come from. However, the inverse-optimal identification of aggregate G-SWF weights only relies on the reduced-form aggregate weights.

The interpretation is that the planner aggregates heterogeneous individual social evaluations, as in the double-aggregation problem emphasized by Roberts (1995). Agents

may hold different views about how social welfare should evaluate allocations, and these views may depend on the idiosyncratic histories they have experienced. As explained in the introduction, survey and experimental assessments of individual social evaluations suggest that such views are heterogeneous and may evolve with life experience. The following construction translates this idea into a reduced-form H-SWF. In a first step, each history type assigns welfare weights to other history types. In a second step, the planner aggregates these social evaluations linearly across the population. The resulting aggregate weights are the history-based weights that enter the H-SWF.

In the two-period economy, consider an agent whose period-1 productivity is x . This agent's period-1 social evaluation assigns weight $\nu_{1,x,y}$ to agents with period-1 productivity y . In period 2, an agent with realized history (x, x') assigns weight $\nu_{2,(x,x'),(y,y')}$ to agents with history (y, y') . The notation therefore records two histories: the history of the evaluating agent and the history of the agents whose welfare is being evaluated.

The planner then aggregates these individual social evaluations. Its aggregation weight on the evaluation of agents with initial productivity x is denoted by p_x . Under linear aggregation, the aggregate history-based weights are:

$$\begin{aligned}\tilde{\omega}_{1,y} &:= \sum_{x \in \mathcal{Y}} \pi_x p_x \nu_{1,x,y}, \\ \tilde{\omega}_{2,y,y'} &:= \sum_{x,x' \in \mathcal{Y}} \pi_x p_x \Pi_{x,x'} \nu_{2,(x,x'),(y,y')}.\end{aligned}$$

Substituting these weights into the aggregate criterion gives an H-SWF of the form introduced above. The H-SWF can therefore be interpreted as the reduced-form outcome of linearly aggregating history-dependent social evaluations across agents.

The timing of these evaluations is not central for what follows. If agents do not anticipate how their social evaluations change, the construction aggregates their realized period-specific evaluations. If agents do anticipate those changes, the same logic can be written with different primitive individual evaluations but delivers the same reduced-form H-SWF weights. We therefore work directly with the reduced-form H-SWF and study two restrictions of it below.

2.2.3 A first restriction: the PN-SWF

The first restriction collapses histories to lifetime identities, which is the initial productivity state in the two-period environment. A PN-SWF therefore assigns a fixed welfare weight to all agents who start from the same initial productivity, independently of the period-2 shock they later receive. This is the standard Pareto-Negishi restriction.

The restriction is most directly stated on the H-SWF weights. For each initial productivity y , all period-2 histories that start from y inherit the same fixed-origin weight.

Thus:

$$\tilde{\omega}_{2,y,y'} = \tilde{\omega}_{1,y} \quad \text{for all } y' \in \mathcal{Y}.$$

The fixed-origin weight is therefore attached to the initial state, not to the realized period-2 state. The resulting welfare criterion is:

$$W^P = \sum_{y \in \mathcal{Y}} \pi_y \tilde{\omega}_{1,y} \left[u(c_{1,y}) + \beta \sum_{y' \in \mathcal{Y}} \Pi_{yy'} u(c_{2,y,y'}) \right]. \quad (3)$$

Heterogeneity in individual social evaluations is not enough to generate current-state welfare weights. It delivers a PN-SWF when those evaluations attach to lifetime identities. The reason is that all future histories descending from the same initial type inherit the same welfare weight. Indeed, if agents with initial productivity x value the lifetime utility V_y of agents with initial productivity y with weight $\alpha_{x,y}$, and if the planner aggregates these views with weights p_x , then the planner SWF is the same as in (3), with induced PN-SWF weight equal to $\tilde{\omega}_{1,y} = \sum_{x \in \mathcal{Y}} \pi_x p_x \alpha_{x,y}$.

This restriction has a direct implication for planner allocations: with complete non-distortionary instruments, the PN-planner insures idiosyncratic shocks within each origin group.

Lemma 1. *Under complete non-distortionary instruments, and the two-period resource constraints, a PN-planner implements an allocation that is constant over time and period-2 realizations, conditional on the initial productivity state: $c_{1,y} = c_{2,y,y'}$ for any $y, y' \in \mathcal{Y}$.*

Lemma 1 records the allocation implication of fixed-lifetime weights. The planner may choose different consumption levels for agents with different initial productivity states, because their PN-SWF weights may differ. But once the initial state is fixed, realized idiosyncratic shocks do not change the welfare weight attached to that origin, and the planner allocates the same consumption to all such histories.

We depart from the PN-SWF for the following reason. As weights are attached to origins, they cannot depend on the period-2 productivity realization y' . This is very restrictive because current productivity has first-order effects on welfare-related outcomes. In the stationary infinite-horizon economy, this restriction becomes even stronger: fixed-lifetime Pareto weights collapse to the utilitarian benchmark (see Section 3.4). Since our goal is to allow for more general SWFs (with the utilitarian SWF as a special case), we depart from the PN-SWF.

The next restriction moves in the opposite direction. Instead of collapsing histories to origins, it collapses histories to contemporaneous individual states. This is the G-SWF restriction used in the empirical part of the paper.

2.2.4 A second restriction: the G-SWF

In the notation of the H-SWF, the G-SWF is the restriction under which period-2 welfare weights depend only on contemporaneous productivity, rather than on the full idiosyncratic history. The contrast with the PN-SWF is therefore direct: the PN-SWF keeps the origin y and ignores the realized period-2 state, whereas the G-SWF keeps the realized period-2 state y' and ignores the origin. Formally, there exist period-2 current-state weights such that the H-SWF weights satisfy:

$$\tilde{\omega}_{2,y,y'} = \tilde{\omega}_{2,y'} \quad \text{for all } y \in \mathcal{Y}.$$

The associated welfare criterion is:

$$\mathcal{W}^G = \sum_{y \in \mathcal{Y}} \pi_y \tilde{\omega}_{1,y} u(c_{1,y}) + \beta \sum_{y,y' \in \mathcal{Y}} \pi_{yy'} \tilde{\omega}_{2,y'} u(c_{2,y,y'}).$$

Under this restriction, histories still matter for allocations, since they still determine transition masses and individual consumption histories. The restriction only applies to the welfare weight attached to a period-2 agent, which must be a function of the agent's contemporaneous observable state. Thus two agents with the same current productivity receive the same period-2 welfare weight, even if they arrived at that state from different period-1 productivity levels. This is why the G-SWF is the object that can be disciplined by cross-sectional distributional data.

This restriction has a different implication from the PN-SWF for planner allocations. With complete non-distortionary instruments, the G-planner pools agents by their current productivity state rather than by their origin.

Lemma 2. *A G-planner implements period-2 current-state pooling: $c_{2,y,y'} = c_{2,z,y'}$ for any $y, z, y' \in \mathcal{Y}$. Period-1 consumption can depend on the period-1 productivity state.*

Lemma 2 is the mirror image of Lemma 1. Once current productivity is fixed, origins no longer affect period-2 consumption. Period-2 consumption can still differ across current productivity states, because G-SWF weights may differ across these states; period-1 consumption can also vary with period-1 productivity.

2.2.5 Inverse-optimal approach using the G-SWF

Lemma 2 gives the forward implication of the G-SWF: if welfare weights depend only on contemporaneous individual states, then the planner pools period-2 consumption across origins conditional on the current state. The inverse-optimal approach uses the same optimality conditions in the opposite direction. Given an observed allocation, it recovers the G-SWF weights that rationalize the allocation as optimal. The recovery depends on two objects: the period utility function and the set of instruments available to the planner.

Proposition 2 (Observable-state recovery). *Suppose the observed two-period allocation is rationalized by a planner with a G-SWF, non-distortionary instruments and strictly positive marginal utilities. Then normalized current-state G-SWF weights are pinned down by marginal-utility ratios. For any states $y, z \in \mathcal{Y}$,*

$$\frac{\tilde{\omega}_{1,y}}{\tilde{\omega}_{1,z}} = \frac{u'(c_{1,z})}{u'(c_{1,y})}, \quad \frac{\tilde{\omega}_{2,y}}{\tilde{\omega}_{2,z}} = \frac{u'(c_{2,z})}{u'(c_{2,y})},$$

where in the second-period ratio, $c_{2,y}$ denotes the common value of $c_{2,x,y}$ across origin states x .

Proposition 2 is a positive identification result. These cross-sectional observable data recover the normalized G-SWF weights that rationalize the allocation. Fixed-lifetime Pareto weights require origin-preserving information, which lies outside the current-state cross section.

This restriction makes welfare weights depend only on variables observed in the cross section. It is therefore the relevant welfare object for the baseline inverse-optimal exercise below. A richer H-SWF can still be approximated in the truncated model of Section 4.5 by assigning separate weights to truncated histories (and not only to current productivity level). In the robustness exercise reported in Appendix F.2, averaging those recovered history weights within current-productivity states yields a profile close to the non-parametric G-SWF profile. We therefore use the G-SWF as our baseline welfare criterion.

Because the recovery uses marginal utilities, the recovered relative weights are invariant to common positive affine normalizations of the period utility function. This invariance is limited to common affine normalizations: arbitrary increasing transformations of period utility or individual intertemporal utility generally change the recovered marginal-utility ratios. The quantitative exercise below therefore treats the utility specification as part of the maintained model and reports robustness across alternative specifications. By contrast, applying an increasing transformation to the planner objective after aggregation leaves the planner's choice unchanged: this is only an ordinal relabeling of the social objective.

We impose a monotonicity requirement on our welfare criterion: aggregate weights must be positive, so lowering any agent's utility cannot increase social welfare.² Since only relative weights are identified, we fix scale by normalizing the two current-state weight

²For general social objectives, Pareto comparisons require care (see Sen, 1970, Kaplow and Shavell, 2001, Fleurbaey and Trannoy, 2003).

vectors with respect to the marginal productivity distribution:

$$\begin{aligned}\tilde{\omega}_{1,y} &> 0, & \tilde{\omega}_{2,y} &> 0, \\ \sum_{y \in \mathcal{Y}} \pi_y \tilde{\omega}_{1,y} &= 1, & \sum_{y \in \mathcal{Y}} \pi_y \tilde{\omega}_{2,y} &= 1.\end{aligned}$$

For period 2, this is equivalent to the normalization under the history distribution because stationarity implies $\sum_{x,y \in \mathcal{Y}} \pi_x \Pi_{xy} \tilde{\omega}_{2,y} = \sum_{y \in \mathcal{Y}} \pi_y \tilde{\omega}_{2,y}$.

Together, the inverse-optimal conditions and normalization complete the two-period illustration: normalized current-state weights are recoverable from observed allocations once period utility and the instrument set are fixed.

Table 1 summarizes the three SWF candidates and the data needed to discipline each of them.

Object	Weights	Needed data	Period-2 pattern
H-SWF	History weights $\omega_{1,y}, \omega_{2,y,y'}$	History-level or transition-cell data	Full-history variation in $c_{2,y,y'}$
PN-SWF	Origin weights $\tilde{\omega}_{1,y}$	Origin-preserving cohort, panel, or transition-cell data	Origin pooling, $c_{2,y,y'} = c_{2,y,z}$
G-SWF	Current-state weights $\tilde{\omega}_{2,y'}$	Cross-sectional data and Ramsey FOCs	Current-state pooling, $c_{2,y,y'} = c_{2,z,y'}$

Table 1: The unrestricted history-based SWF and its two low-dimensional restrictions.

3 Construction of the G-SWF in the intertemporal case

This section carries the previous construction from the finite illustration into the stationary Bewley environment used in the quantitative part of the paper. We first define histories and intertemporal welfare. We then construct the unrestricted history-based SWF. Next, we restrict welfare weights to depend only on variables observed in the stationary cross section to obtain the G-SWF used in the Ramsey inverse-optimal exercise. We close by showing that a fixed-lifetime Pareto planner collapses to a utilitarian planner in a stationary economy.³

³Some studies (e.g., Chang et al., 2018) allow social weights to depend on endogenous variables, such as consumption or wealth. This creates a feedback from allocations to the planner's objective: when social weights vary with the allocation, the planner must internalize how policy changes its own weights. This can create fixed-point and multiplicity issues. We therefore let social weights depend on

3.1 The setup

We consider a stationary infinite-horizon economy with incomplete financial markets. The full Ramsey model below adds assets, production, and fiscal instruments. Here we isolate the history structure needed to define the planner SWFs. Time is discrete, indexed by $t \geq 0$. The population consists of a continuum $\mathcal{I}$ of ex-ante identical agents of measure one.

Risk structure. Idiosyncratic risk is modeled as an uninsurable individual productivity shock y_t that can take Y distinct values in the finite set $\mathcal{Y}$. The productivity risk follows a first-order Markov chain with transition matrix $(\Pi_{yy'})_{y,y' \in \mathcal{Y}}$. This matrix is assumed to be irreducible and aperiodic, ensuring a unique stationary distribution $(\pi_y)_{y \in \mathcal{Y}}$, satisfying $\sum_{y \in \mathcal{Y}} \pi_y = 1$.

Let $y^t = \{\dots, y_{t-1}^t, y_t^t\}$ denote a one-sided infinite sequence of productivity levels representing an agent's history up to date t . The set of all such histories is $\mathcal{Y}^\infty$. Since we track the evolution of histories over time, we retain time subscripts for clarity. For a given history $y^t \in \mathcal{Y}^\infty$, we use: (i) $y_\tau^t \in \mathcal{Y}$ for the productivity level at date $\tau \leq t$; (ii) $y^{s,t}$ for the truncation of y^t at date $s \leq t$, such that $y^{s,t}$ and y^t coincide for all $\tau \leq s$, i.e., $y_\tau^{s,t} = y_\tau^t$. Distinct histories are indicated with decorators, e.g., $\tilde{y}^t$ and y^t .

Initial distribution. To start directly from a stationary cross-section, we impose two assumptions on the initial distribution: (i) all agents start the economy with an initial infinite history in $\mathcal{Y}^\infty$; (ii) initial wealth depends only on an agent's initial history. This assumption encompasses, as special cases, economies in which all agents share identical wealth or start from the steady-state wealth distribution. Because our analysis focuses on stationary allocations and distributions, this restriction keeps the welfare construction focused on histories rather than on transitory variation in the initial wealth distribution.

The sequential representation. Because the next subsection constructs a history-based SWF, we need a probability space over histories. In an infinite-horizon model, the set of histories has the cardinality of the continuum (neither finite nor countable). Hence, the probability space must be defined with a general measure. We construct the probability space $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$, where $\mathcal{F}$ is a suitable σ -algebra and μ a measure (see Appendix A.3). For any measurable set $B \in \mathcal{F}$, $\mu(B) \geq 0$ is the measure of agents currently in histories $y^t \in B$.⁴ Normalizing population size to one implies $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$.⁵

idiosyncratic histories only rather than on wealth or other endogenous variables. This choice preserves a clear separation between individual choices and social valuation.

⁴A single history is an event of measure zero in $\mathcal{F}$. Therefore, statements that hold for all histories y^t should be understood as holding almost surely.

⁵In finite time, we would write $\sum_{y^t \in \mathcal{Y}^t} \mu_t(y^t) = 1$ instead of $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$. The finite-time intuition carries over directly to the infinite-horizon framework.

Transitions across histories are defined as follows. Consider $y^{t+1}, \tilde{y}^t \in \mathcal{Y}^\infty$. Denote by $y^{t,t+1}$ the history y^{t+1} up to period t . The probability of moving from $\tilde{y}^t$ in the current period to y^{t+1} in the next period equals the probability of switching from $\tilde{y}_t^t$ to y_{t+1}^{t+1} if $y^{t,t+1}$ and $\tilde{y}^t$ coincide, and zero otherwise. Denote this conditional probability by $\mu_1(\cdot | \tilde{y}^t)$, formally: $\mu_1(dy^{t+1} | \tilde{y}^t) = \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(dy^{t,t+1})$, where $\delta_{\tilde{y}^t}$ is the Dirac delta at $\tilde{y}^t$.⁶

By induction, the probability of moving from $\tilde{y}^t$ to a future history y^{t+s} in $s \geq 1$ periods is $\mu_s(dy^{t+s} | \tilde{y}^t) = \prod_{k=0}^{s-1} \Pi_{y_{t+k}^{t+s} y_{t+k+1}^{t+s}} \times \delta_{\tilde{y}^t}(dy^{t,t+s})$. In words, transitioning from $\tilde{y}^t$ to y^{t+s} requires that the histories coincide up to period t , followed by the cumulative probability of the sequence $y_{t+1}^{t+s}, \dots, y_{t+s}^{t+s}$.

Individual intertemporal welfare. For a given allocation, let $U(y^t)$ denote the period utility of an agent with history y^t . To lighten notation, we suppress explicit dependence on the allocation. In the benchmark case with private consumption and labor supply (as in our quantitative application), $U(y^t) := u(c(y^t)) - v(l(y^t))$, where $c : \mathcal{Y}^\infty \rightarrow \mathbb{R}^+$ and $l : \mathcal{Y}^\infty \rightarrow \mathbb{R}^+$ are the policy functions for consumption and labor.

Intertemporal welfare is time separable and of the expected-utility form. It is thus defined as the discounted sum of expected period utilities over all future dates:

$$V(y^t) = \sum_{s=0}^{\infty} \beta^s \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} U(\tilde{y}^{t+s}) \mu_s(d\tilde{y}^{t+s} | y^t). \quad (4)$$

As we consider a steady-state allocation, intertemporal welfare can be written recursively as:

$$V(y^t) = U(y^t) + \beta \mathbb{E}_{\tilde{y}^{t+1}} [V(\tilde{y}^{t+1}) | y^t], \quad (5)$$

where $\mathbb{E}_{\tilde{y}^{t+1}} [V(\tilde{y}^{t+1}) | y^t] = \int_{\tilde{y}^{t+1} \in \mathcal{Y}^\infty} V(\tilde{y}^{t+1}) \mu_1(d\tilde{y}^{t+1} | y^t)$ is a conditional expectation. See Appendix A.5. These definitions give us the histories, transition probabilities, and continuation utilities needed to define a history-based SWF in the next subsection.

3.2 Constructing the H-SWF

This subsection carries the finite two-period H-SWF into the stationary Bewley environment. In the finite two-period model of Section 2, the H-SWF assigns weights to finite history realizations. In the stationary setting, those realizations become measurable sets of histories. Under the same maintained ingredients, a common cardinal utility scale, monotonicity, continuity, and stable separable trade-offs across history realizations, the resulting SWF is again a history-based additive criterion.

The H-SWF is the reduced-form welfare criterion used below. The empirical exercise

⁶The measure μ_1 satisfies the standard properties of a conditional probability: $\mu_1 \geq 0$, $\int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_1(dy^{t+1} | \tilde{y}^t) = 1$, and $\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(dy^{t+1} | \tilde{y}^t) \mu(d\tilde{y}^t) = \mu(dy^{t+1})$. See Appendix A.4.

later recovers aggregate G-SWF weights after restricting welfare weights to depend only on variables observed in the stationary cross section. The same cardinality caveat carries over from the finite model: the criterion is invariant to common positive affine normalizations of period utility, and hence to the induced affine normalization of continuation utility, but not to arbitrary increasing transformations of individual utility. Section 6 discusses the assumptions in our construction (in particular cardinal comparability) and their robustness.

As in the simple model, we give an interpretation of this H-SWF construction in three steps.

Step 1: Subjective valuation of another history. Consider two agents characterized by their histories $y^t \in \mathcal{Y}^\infty$ and $\tilde{y}^t \in \mathcal{Y}^\infty$ at date t ; the allocation is taken as given. Under our preference specification, the $\tilde{y}^t$ -agent values the allocation associated with any future history $\tilde{y}^{t+s}$ by its period utility $U(\tilde{y}^{t+s})$. The y^t -agent may perceive the same future allocation differently. Specifically, the y^t -agent assigns to the utility $U(\tilde{y}^{t+s})$ of the $\tilde{y}^t$ -agents a corrective factor $\hat{\omega}(y^t, \tilde{y}^{t+s})$, such that $\hat{\omega}(y^t, \tilde{y}^{t+s})U(\tilde{y}^{t+s})$ represents the valuation of the history $\tilde{y}^{t+s}$ by the y^t -agent.⁷

Summing these weighted valuations $\hat{\omega}(y^t, \tilde{y}^{t+s})U(\tilde{y}^{t+s})$ over all future periods and histories – using the appropriate conditional probabilities – yields the total valuation that the y^t -agent assigns to the $\tilde{y}^t$ -agent's lifetime utility. Denoting this by $\hat{V}(y^t, \tilde{y}^t)$, we obtain:

$$\hat{V}(y^t, \tilde{y}^t) = \sum_{s=0}^{\infty} \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \tilde{y}^{t+s}) U(\tilde{y}^{t+s}) \mu_s(d\tilde{y}^{t+s} | \tilde{y}^t). \quad (6)$$

This expression mirrors the definition of the intertemporal utility $V(\tilde{y}^t)$ but includes the weighting terms $\hat{\omega}(y^t, \tilde{y}^{t+s})$. In words, $\hat{V}(y^t, \tilde{y}^t)$ represents the expected intertemporal welfare of an agent currently in history $\tilde{y}^t$, as perceived by another agent whose current history is y^t . Consequently, two agents with different histories today, such as y^t and $y^{*,t}$, can disagree on the welfare of agents in history $\tilde{y}^t$.

This construction embeds an assumption of stable moral values. Agents evaluate the future using the values attached to their current history y^t , despite their own future history evolution. However, this assumption is not essential for our representation: if agents anticipate future changes in social evaluations, the same reduced-form H-SWF can be recovered with expected future weights.

Step 2: Individual social valuation. The individual social valuation for agents with history y^t is obtained by aggregating their subjective valuations $\hat{V}(y^t, \tilde{y}^t)$ over possible

⁷This corrective term parsimoniously captures habit formation or preference heterogeneity: agents with distinct histories may value the same consumption bundle or labor differently.

evaluated histories $\tilde{y}^t$:

$$\mathcal{W}^I(y^t) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t) \mu(d\tilde{y}^t). \quad (7)$$

This object records the individual social evaluation of agents with history y^t , that is, how these agents think the welfare of all others should be accounted for by the planner. As in the simple model, no weight normalization is imposed at this stage.

Step 3: Aggregation into H-SWF. We assume that the planner observes these individual social valuations in the population and aggregates them according to planner aggregation weights assigned to each evaluator history. Agents need not have the same importance for the planner. Formally, the individual social valuation of agents with history y^t receives planner aggregation weight $p(y^t)$, which captures how their social valuation is accounted for by the planner. The H-SWF $\mathcal{W}^H$ thus equals this weighted aggregate social valuation:

$$\mathcal{W}^H = \int_{y^t \in \mathcal{Y}^\infty} p(y^t) \mathcal{W}^I(y^t) \mu(dy^t). \quad (8)$$

Special cases. We illustrate the construction with two special cases. First, we recover the utilitarian H-SWF when all agents assign a constant relative weight across evaluated histories. Formally, if $\hat{\omega}(y^t, \tilde{y}^{t+s}) := \hat{\omega}(y^t)$ for all $\tilde{y}^{t+s}$ the aggregate H-SWF is proportional to the utilitarian SWF: $\mathcal{W}^H = \left(\int_{\tilde{y}^t \in \mathcal{Y}^\infty} p(\tilde{y}^t) \hat{\omega}(\tilde{y}^t) \mu(d\tilde{y}^t) \right) \int_{y^t \in \mathcal{Y}^\infty} V(y^t) \mu(dy^t)$.⁸ Second, we recover the PN-SWF, in which agents are self-interested and care only about histories that they can possibly experience. Formally, $\hat{\omega}(y^t, \tilde{y}^{t+s}) := \delta_{y^t}(\tilde{y}^{t,t+s}) \hat{\omega}(y^t)$, where $\delta_{y^t}(\tilde{y}^{t,t+s}) = 1$ if $\tilde{y}^{t,t+s}$ is compatible with y^t and 0 otherwise. Then the H-SWF is a weighted sum of individual intertemporal utilities: $\mathcal{W}^H = \int_{y^t \in \mathcal{Y}^\infty} p(y^t) \hat{\omega}(y^t) V(y^t) \mu(dy^t)$.

An explicit expression of the H-SWF. We state the following proposition.

Proposition 3. *The H-SWF (8) admits the following expression:*

$$\mathcal{W}^H = \sum_{t=0}^{\infty} \int_{y^t \in \mathcal{Y}^\infty} \beta^t \omega(y^t) U(y^t) \mu(dy^t), \quad (9)$$

where the weights ω are given by:

$$\omega(y^{t+s}) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} p(\tilde{y}^t) \hat{\omega}(\tilde{y}^t, y^{t+s}) \mu(d\tilde{y}^t). \quad (10)$$

Proposition 3 shows that the H-SWF can be written as the discounted sum of period utilities with history-dependent weights ω . In other words, the H-SWF can be seen as a

⁸This property is known at least since Aiyagari and McGrattan (1998) as a justification for using the utilitarian SWF under the veil of ignorance.

generalization of the utilitarian SWF. The utilitarian case obtains when these weights are constant across histories. More generally, the weight $\omega(y^{t+s})$ in (10) can be interpreted as the *average* weight assigned to history y^{t+s} by all agents, where agents are themselves weighted by the planner aggregation weight p . See Appendix A.6 for the details of the proof.

The sequential representation (9) also admits a recursive form:

$$\mathcal{W}^H = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t) \mu(dy^t) + \beta \cdot \mathcal{W}^H,$$

which extends the recursive representation of the utilitarian H-SWF to history-dependent weights. Under aggregate dynamics, period utility U_t is time dependent (through the allocation). The H-SWF then satisfies $\mathcal{W}_t^H = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U_t(y^t) \mu(dy^t) + \beta \cdot \mathcal{W}_{t+1}^H$. This representation remains too high-dimensional for empirical work because the weights may depend on full histories. The next subsection restricts welfare weights to depend only on variables observed in the stationary cross section, yielding the G-SWF used in the Ramsey inverse-optimal exercise.

3.3 Constructing the G-SWF

The H-SWF expression in Proposition 3 is fully general but not immediately suited for estimation, owing to its high-dimensionality. We therefore introduce a tractability assumption that yields G-SWF weights. We assume that agents with the same current productivity assign the same value to future histories, and that this value depends only on the last productivity of the future history under consideration. We also restrict planner aggregation weights to depend only on current productivity. In words, the G-SWF restricts welfare weights to depend only on variables observed in the stationary cross section. This restriction makes the G-SWF recoverable from stationary cross-sectional data. We check that more general specifications deliver similar quantitative results (see Section 6).

Assumption A. *There exist two functions, denoted (with slight abuse) $\tilde{\omega} : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ and $p : \mathcal{Y} \rightarrow \mathbb{R}$ such that for all histories $y^t, \tilde{y}^{t+s} \in \mathcal{Y}^\infty$, we have:*

$$\begin{cases} \tilde{\omega}(y^t, \tilde{y}^{t+s}) := \hat{\omega}_{y_t^t, \tilde{y}_{t+s}^{t+s}}, \\ p(y^t) := p_{y_t^t}, \end{cases}$$

where we recall that y_t^t is the current productivity level in history y^t .

Assumption A reduces dimensionality by defining weights as a finite vector indexed by current productivity. Under this restriction, all histories ending in productivity state y

receive the same aggregate G-SWF weight, denoted ω_y and defined for all $y \in \mathcal{Y}$ as:

$$\omega_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}} p_{\tilde{y}} \tilde{\omega}_{\tilde{y},y},$$

where $\pi_{\tilde{y}}$ is the share of agents with productivity $\tilde{y}$. We normalize the weights with $\sum_y \pi_y \omega_y = 1$. The monotonicity restriction is imposed directly on the G-SWF weights: recovered weights must be nonnegative in every productivity state. The functions p and $\hat{\omega}$ in Assumption A are one interpretation of these aggregate weights but are not separately identified.

Identification of G-SWF weights. We identify the G-SWF weights by finding $(\omega_y)_{y \in \mathcal{Y}}$ such that a given fiscal system (typically the observed one) is optimal in the Ramsey problem. Formally, the FOCs of the Ramsey planner imply linear constraints for the SWF weights $(\omega_y)_{y \in \mathcal{Y}}$. The number of constraints, n , depends on the available planner's instruments. In general, the system (formed by the n linear constraints plus the normalization) is underdetermined because the number of weights exceeds $n + 1$.⁹

Following Heathcote and Tsujiyama (2021), we resolve underdetermination by assuming a parametric specification for ω_y with exactly $n + 1$ parameters. Under mild conditions on the functional form, the weights are then identified uniquely as the solution to a nonlinear system. We formalize this as follows.

Definition 1. Given $1 \leq n \leq |\mathcal{Y}| - 1$ linear constraints represented by the matrix $(L_{k,y})_{k=1,\dots,n; y \in \mathcal{Y}}$ and a set of parametric functions $f_y : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ ($y \in \mathcal{Y}$) for the weights, the estimated SWF weights are $(\omega_y)_{y \in \mathcal{Y}} = (f_y(\theta))_y$, where $\theta \in \mathbb{R}^p$ solves:

$$0 = \sum_{y \in \mathcal{Y}} L_{k,y} f_y(\theta) \text{ for all } k = 1, \dots, n, \quad (11)$$

$$1 = \sum_{y \in \mathcal{Y}} \pi_y f_y(\theta). \quad (12)$$

As a robustness check, we also consider a non-parametric estimation. Without the log-quadratic restriction, the system (11)–(12) leaves the weights underdetermined. We therefore select the solution with the lowest variance across productivity levels. Both parametric and non-parametric approaches deliver quantitatively similar weights. This exercise adds support to our estimation strategy. See Appendix F.2 for definitions and results in the non-parametric case.

⁹In our quantitative exercise, there are 10 idiosyncratic states and 4 planner instruments.

3.4 Why the fixed-lifetime PN-SWF converges to the utilitarian SWF

Having constructed the H-SWF and its G-SWF restriction, we now clarify the properties of fixed-lifetime Pareto weights in a stationary Bewley economy. Starting from any distribution of Pareto weights, the Markovian dynamics repeatedly reshuffle identities across current asset-productivity states. At the stationary steady state, any Pareto-weighted distribution converges to the same invariant distribution. Hence, fixed-lifetime Pareto weights converge to the utilitarian benchmark. The result below formalizes this convergence.

To state the result, let an individual current state be $(a, y) \in \mathcal{A} \times \mathcal{Y}$, where a denotes assets in the asset space $\mathcal{A}$ and y denotes productivity. Let $\Pi = (\Pi_{yy'})$ denote the transition matrix for idiosyncratic productivity. A stationary equilibrium is summarized by policy functions, including the consumption policy function $c = c(a, y)$. Together with Π , these policy functions define the policy-induced transition over current asset-productivity states and hence induce a Markov process $((a_{i,t}, y_{i,t}))_{t \geq 0}$ over current states. Define the stationary continuation value:

$$\mathcal{V}(a, y) = \mathbb{E}_{a,y} \left[\sum_{s=0}^{\infty} \beta^s u(c(a_s, y_s)) \right].$$

A PN-SWF with fixed Pareto weights $(\omega_i)_{i \in \mathcal{I}}$ satisfying $\omega_i \geq 0$ and $\int_{\mathcal{I}} \omega_i di = 1$ evaluates date- t welfare as:

$$W_t^P = \int_{\mathcal{I}} \omega_i \mathcal{V}(a_{i,t}, y_{i,t}) di.$$

The weights ω_i are attached to lifetime identities, not to the current realizations $(a_{i,t}, y_{i,t})$.

Under standard conditions, the stationary policy functions and shock process induce a common ergodic Markov process on current states (a, y) : for any initial distribution over current states, the distribution converges to the unique invariant distribution μ^* .¹⁰ See, for example, Huggett (1993). We focus on the implication for fixed-lifetime Pareto weights.

Proposition 4 (Ergodicity and the utilitarian limit). *Under this common-ergodicity assumption, for any fixed Pareto weights satisfying the normalization above, the fixed-lifetime PN-SWF converges to the utilitarian limit:*¹¹

$$\lim_{t \rightarrow \infty} W_t^P = \frac{1}{1 - \beta} \int_{\mathcal{A} \times \mathcal{Y}} u(c(a, y)) \mu^*(da, dy).$$

¹⁰In finite-grid implementations, a sufficient condition is that the induced transition matrix over joint asset-productivity states is irreducible and aperiodic.

¹¹In finite-grid implementations, $\mathcal{V}$ is bounded, so convergence of the induced distribution implies convergence of $\int \mathcal{V} d\nu_t^\omega$. In the general state-space case, the statement requires this same integral convergence; a sufficient condition is uniform integrability of $|\mathcal{V}|$ under the Pareto-weighted laws $(\nu_t^\omega)_t$.

The limit is independent of the Pareto weights (ω_i) .

Proposition 4 shows that fixed-lifetime Pareto weights disappear in the stationary representation. Pareto weights can matter along transition paths because they change the weighted distribution of identities across current states. Under common ergodicity, however, all such weighted distributions converge to the same invariant distribution. Hence the stationary limit of the PN-SWF is the utilitarian SWF under μ^* .

These considerations make the G-SWF the relevant stationary inverse-optimal welfare object. The convergence result shows that fixed-lifetime Pareto weights do not deliver a non-utilitarian stationary representation in the stationary Bewley economy. The PN-SWF is therefore not well-suited for the stationary inverse-optimal exercise. The same issue arises with finite lifetimes, when Pareto-Negishi weights are assigned at birth and the data omit origin or cohort labels. In the absence of origin-preserving information, a non-utilitarian stationary welfare object must indeed attach weights to current states.

4 The Ramsey program and the inverse-optimal approach

This section embeds the G-SWF welfare object constructed in Section 3 in a quantitative Ramsey economy. Evaluating the Ramsey first-order conditions at the observed fiscal system and stationary distribution yields restrictions on the G-SWF weights. These restrictions are not sufficient by themselves to identify the weights, which also requires a normalization and, in the baseline, a parameterization of the weight profile.

The economy features a unit mass of ex-ante identical agents facing idiosyncratic productivity risk y , with the same risk structure as in Section 3. A representative firm produces a final consumption good using capital and efficient labor. We first describe production, fiscal instruments, households, and competitive equilibrium. We then write the Ramsey program and show how the Ramsey steady-state first-order conditions restrict the G-SWF weights.

4.1 Production and government

Production. In each period t , a production technology with constant returns to scale transforms capital K_{t-1} and labor L_t into $F(K_{t-1}, L_t)$ units of output. The production function is smooth in K and L and satisfies the standard Inada conditions. Depreciation is subsumed into F . Labor L_t is measured in efficiency units. Output is produced by a representative profit-maximizing firm. Let $\tilde{w}_t$ denote the real before-tax wage and $\tilde{r}_t$ the

real before-tax rental rate of capital. Profit maximization implies, for $t \geq 1$,

$$\tilde{r}_t = F_K(K_{t-1}, L_t) \quad \text{and} \quad \tilde{w}_t = F_L(K_{t-1}, L_t). \quad (13)$$

Government. The government finances a path of public spending (G_t) with one-period default-free debt and distortionary taxes on consumption, on labor income, and on capital income. With no aggregate risk, public debt and capital are perfect substitutes and earn the same pre-tax return $\tilde{r}_t$. Consumption and capital taxes are linear and denoted by τ_t^c and τ_t^K . Labor income taxation follows the functional form of Heathcote et al. (2017) (HSV, henceforth). If an agent with productivity y supplies labor l , pre-tax labor income is $\tilde{w}_t y l$, the labor income tax, $\mathcal{T}_t$, is:

$$\mathcal{T}_t(\tilde{w}_t y l) := \tilde{w}_t y l - \kappa_t (\tilde{w}_t y l)^{1-\tau_t}, \quad (14)$$

where κ_t controls the level of taxation and τ_t its progressivity – both are planner instruments. When $\tau_t = 0$, labor tax is linear with rate $1 - \kappa_t$. When $\tau_t = 1$, all agents earn the same post-tax income κ_t . This functional form, together with a linear capital tax, reproduces salient features of the US and French tax systems and their progressivity.¹²

Government revenues are the sum of consumption-tax receipts, labor-income-tax receipts, and capital-income-tax receipts. Formally, they are equal to $\tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^K \tilde{r}_t (K_{t-1} + B_{t-1})$, where C_t is aggregate consumption and $A_{t-1} := K_{t-1} + B_{t-1}$ denotes aggregate savings at $t - 1$. The government budget constraint is:

$$G + (1 + \tilde{r}_t) B_{t-1} = \tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^K \tilde{r}_t A_{t-1} + B_t. \quad (15)$$

It is useful to rewrite this constraint in post-tax rates r_t and w_t , defined as follows:

$$r_t := (1 - \tau_t^K) \tilde{r}_t, \quad w_t := \kappa_t (\tilde{w}_t)^{1-\tau_t}. \quad (16)$$

Using constant returns-to-scale and (16), the government budget constraint (15) becomes:

$$G + (1 + r_t) B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t) + B_t. \quad (17)$$

4.2 Households' program

Period utility. Agents have separable period utility over private consumption and labor:

$$U(c, l) := u(c) - v(l). \quad (18)$$

¹²The literature uses either the combination of a linear tax and a lump-sum transfer (e.g., Dyrda and Pedroni, 2022; Açıkgöz et al., 2022) or the HSV structure (see Ferriere and Navarro, 2023). Heathcote and Tsujiyama (2021) show that the HSV structure is quantitatively more relevant for capturing observed progressivity.

The function $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly concave, with $u'(0) = \infty$, while $v : \mathbb{R}_+ \rightarrow \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly convex, with $v'(0) = 0$.

Agents' program. Agents' resources consist of labor income and savings payoffs. For agent i with productivity $y_{i,t}$ and labor $l_{i,t}$, post-tax labor income is $\tilde{w}_t y_{i,t} l_{i,t} - \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) = w_t (y_{i,t} l_{i,t})^{1-\tau_t}$. Since public debt and capital are perfect substitutes, savings pay $(1 + r_t) a_{i,t-1}$. The agent chooses consumption, labor, and next-period assets to solve:

$$\max_{\{c_{i,t}, l_{i,t}, a_{i,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (u(c_{i,t}) - v(l_{i,t})), \quad (19)$$

$$(1 + \tau_t^c) c_{i,t} + a_{i,t} \leq w_t (y_{i,t} l_{i,t})^{1-\tau_t} + (1 + r_t) a_{i,t-1}, \quad (20)$$

$$a_{i,t} \geq -\bar{a}, c_{i,t} > 0, l_{i,t} > 0, \quad (21)$$

where $\mathbb{E}_0$ is an expectation operator over idiosyncratic risk, and where the initial state $(y_{i,0}, a_{i,-1})$ is given.

The agents' decisions depend on their initial asset $a_{i,-1}$ and their idiosyncratic history y_i^t . To simplify notation, we write optimal choices with individual subscripts (i, t) rather than their full dependence on histories as in Section 3. Aggregates are then obtained by integrating individual choices over the population measure: $\int_i X_{i,t} \ell(di)$, for any variable $X_{i,t}$.

The FOCs associated with the agents' program (19)–(21) are:

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1 + r_{t+1}) \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (22)$$

$$v'(l_{i,t}) = \frac{1 - \tau_t}{1 + \tau_t^c} w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (23)$$

where $\beta^t \nu_{i,t}$ is the Lagrange multiplier on agent i 's credit constraint at t .

Market clearing and resource constraints. The clearing conditions for capital and labor markets can be written as follows:

$$\int_i a_{i,t} \ell(di) = A_t = B_t + K_t, \quad \int_i y_{i,t} l_{i,t} \ell(di) = L_t. \quad (24)$$

Equilibrium definition. Given a fiscal policy, a competitive equilibrium is a collection of individual decisions, of aggregate quantities, and of prices such that agents optimize, markets clear, the government budget is balanced and factor prices are consistent with firm optimality and the tax system. A formal definition is in Appendix B. A steady-state competitive equilibrium is such that prices, allocations, and aggregates are time invariant.

4.3 The Ramsey problem and the identification of weights

The Ramsey planner evaluates allocations with the G-SWF. The welfare weights are functions of the current productivity state observed in the stationary data. With separable period utility, the date-0 Ramsey objective is:

$$\mathcal{W}_0^G := \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{y_{i,t}} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \right], \quad (25)$$

where $(\omega_y)_{y \in \mathcal{Y}}$ is the current-productivity G-SWF weight vector. This SWF assigns the same social marginal valuation to agents with the same current productivity, even though they differ in histories or assets.

The Ramsey planner chooses a fiscal policy and the associated competitive-equilibrium allocation that maximize the G-SWF (25) subject to the government budget constraint. Formally,

$$\max_{(w_t, r_t, \tilde{w}_t, \tilde{r}_t, \tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t, G_t, K_t, L_t, (c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_i)_{t \geq 0}} \mathcal{W}_0^G, \quad (26)$$

$$G_t + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t, s_t) + B_t, \quad (27)$$

$$\text{for all } i \in \mathcal{I}: (1 + \tau_t^c) c_{i,t} + a_{i,t} = (1 + r_t) a_{i,t-1} + w_t (y_{i,t} l_{i,t})^{1-\tau_t}, \quad (28)$$

$$a_{i,t} \geq -\bar{a}, \quad \nu_{i,t}(a_{i,t} + \bar{a}) = 0, \quad \nu_{i,t} \geq 0, \quad (29)$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1 + r_{t+1}) \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (30)$$

$$v'(l_{i,t}) = \frac{(1 - \tau_t)}{1 + \tau_t^c} w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (31)$$

$$K_t + B_t = \int_i a_{i,t} \ell(di), \quad L_t = \int_i y_{i,t} l_{i,t} \ell(di), \quad (32)$$

$$\tilde{r}_t = F_K(K_{t-1}, L_t) \quad \text{and} \quad \tilde{w}_t = F_L(K_{t-1}, L_t), \quad (33)$$

$$r_t = (1 - \tau_t^K) \tilde{r}_t, \quad w_t = \kappa_t (\tilde{w}_t)^{1-\tau_t}. \quad (34)$$

plus positivity of c and l and the given initial conditions.

The Ramsey program internalizes the competitive-equilibrium restrictions. Hence, individual budget constraints, borrowing constraints, Euler equations, and labor-supply FOCs all enter as constraints on the planner's choice set. Moreover, the fiscal policy must also satisfy the government budget constraint (27).

The Ramsey program can be simplified before implementing the inverse-optimal approach. First, with a linear capital tax, a progressive labor tax, and one-period public debt, consumption taxes are redundant. Second, following Chamley (1986), we can reformulate the problem in post-tax prices. The planner's instruments are post-tax wage and interest rate (W_t, R_t) , labor tax progressivity, and public debt. The other choice variables are individual and aggregate allocations satisfying competitive-equilibrium

conditions. Proposition 5 in Appendix C.1 states the reformulated program.

A *Ramsey allocation* solves (26)–(34). A *Ramsey steady state* is a Ramsey allocation with constant prices and aggregates. The inverse-optimal step uses the Ramsey first-order conditions in reverse. Evaluated at the observed steady-state fiscal system and stationary distribution, these conditions imply linear restrictions on the G-SWF weights. The next subsection explains which restrictions are available and how the parametric and non-parametric exercises close the recovery problem.

4.4 Expression of the G-SWF

At the steady state, the Ramsey first-order conditions do not identify G-SWF weights. Indeed, consumption taxes are redundant in the reformulated Ramsey program. The public-debt first-order condition pins down the steady-state before-tax real interest rate $1 + \tilde{r} = 1/\beta$ but does not restrict the social weights. The remaining instruments (τ^K, κ, τ) generate three first-order conditions: one determines the multiplier μ_t on the government budget constraint, and two provide independent linear restrictions on the G-SWF weights.

The unrestricted vector $(\omega_y)_{y \in \mathcal{Y}}$ is therefore underidentified by the Ramsey FOCs and normalization. The baseline case closes the system by imposing a log-quadratic profile in productivity, which extends the parametric form in Heathcote and Tsujiyama (2021):

$$\forall y \in \mathcal{Y}, \log \omega_y := \bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2, \quad (35)$$

where $(\bar{\omega}_i)_{i=0,1,2}$ are free parameters. As a robustness check, the non-parametric exercise in Appendix F.1 drops this functional form and selects, among the weight vectors satisfying the same Ramsey restrictions, the profile closest to the utilitarian benchmark.

4.5 Solution method and algorithm

Solving the Ramsey problem directly is difficult because it involves a joint distribution of wealth and Lagrange multipliers. We therefore rely on the truncation method, developed in recent work (LeGrand and Ragot, 2022b, 2023, 2025). Relative to that work, we extend the approach to recover G-SWF weights with a utility function separable in consumption and labor, instead of the GHH specification considered in previous papers.

The method constructs a finite state-space approximation, derives the Ramsey FOCs in that environment, and then applies the inverse-optimal approach to recover the G-SWF weights. The truncated histories are a computational device for approximating the model equations and Ramsey FOCs. The solution method has the following steps.

1. Simulate the heterogeneous-agent model at the steady state under realistic fiscal instruments and income-wealth inequality. Standard techniques deliver the steady-state wealth distribution $\Lambda(a, y)$ for each idiosyncratic state $y \in \mathcal{Y}$ and asset holding

- a , and policy rules for assets, consumption, and labor, $g_a(a, y)$, $g_c(a, y)$, and $g_l(a, y)$.
2. Fix a finite set $\mathcal{H}$ of histories with Markovian transitions. A natural choice is the set of all histories of length N , yielding Y^N truncated histories when there are Y idiosyncratic states.
 3. For a truncated history $y^N := \{y_1, \dots, y_N\} \in \mathcal{H}$, representing agents who experienced y^N over the last N periods, use $\Lambda(\cdot, y_1)$ and the policy rules to compute the wealth distribution $\tilde{\Lambda}(\cdot, y^N)$ for the truncated histories.
 4. Aggregate over $\tilde{\Lambda}$ to construct truncated-history equations. The size of y^N is $S_{y^N} = \int_0^\infty \tilde{\Lambda}(da, y^N)$; per-capita consumption is $c_{y^N} = \int_0^\infty g_c(a, y^N) \tilde{\Lambda}(da, y^N) / S_{y^N}$; average marginal utility satisfies $\int_0^\infty u'(g_c(a, y^N)) \tilde{\Lambda}(da, y^N) / S_{y^N} := \xi_{y^N}^u u'(c_{y^N})$, where $\xi_{y^N}^u$ captures nonlinearity and within- y^N wealth heterogeneity. This yields Y^N budget constraints, Euler equations, and labor choices (see equations (82)–(84) in Appendix D.1). This defines the *truncated model*.
 5. Derive the FOCs of the planner in the truncated model (see Appendix D.2).
 6. Extract from those FOCs two linear constraints on the G-SWF weights and feed them into (11) within Definition 1.
 7. Use the parametric form $\omega_y = e^{\bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2}$ with parameters $(\bar{\omega}_i)_{i=0,1,2}$ and apply the identification strategy in Definition 1. This is a direct generalization of Heathcote and Tsujiyama (2021)’s parametric form, expanding from two to three degrees of freedom.

The detailed derivations for each step are in Appendix D. In the baseline implementation, we consider 10 idiosyncratic states and set the truncation length to $N = 5$, which gives 10^5 possible histories. The recovery is computationally light: the baseline estimation runs in a few minutes, and the recovered weights are stable when the truncation length is increased.

5 Recovering G-SWF weights

The quantitative exercise first calibrates the US and French economies to match their fiscal systems, wealth distributions, and key aggregate ratios. It then recovers the G-SWF weights that rationalize each observed fiscal system under the maintained Ramsey model. The baseline recovery uses the log-quadratic specification introduced above, while non-parametric weights and the preference-parameter exercises in Section 6.2 assess robustness.

5.1 The French and US fiscal systems

We document the tax systems of the United States and France, focusing on average values from 1995 to 2007. This window precedes the Great Recession and the COVID-19 period, which generated large transitory fiscal responses.¹³ We construct country-level averages using Trabandt and Uhlig (2011), reported in Table 2, which also includes inequality measures.

	Total taxes (% GDP)	τ^K (%)	τ^L (%)	τ^c (%)	B (% GDP)	G	Gini before redist.	Gini after redist.	Gini wealth
France	40	35	46	18	60	24	0.48	0.28	0.68
United States	26	36	28	5	63	15	0.48	0.40	0.77

Table 2: Summary of fiscal systems and inequalities in the United States and in France. Total taxes, public debt B and public spending G in percentage of GDP; tax rates τ^K , τ^L and τ^c in percent; Gini indices are unitless.

The first column reports the total mandatory tax revenues as a share of GDP for the two countries. Following the literature, we decompose total levies into three implicit tax rates: capital tax (τ^K), labor tax (τ^L), and consumption tax (τ^c). Columns 2–4 report these rates, computed, respectively, as capital tax receipts divided by the capital stock, labor tax receipts divided by aggregate labor income, and consumption tax receipts divided by consumption. Since the work of Mendoza et al. (1994), this decomposition has been widely used for cross-country comparisons, including by the OECD and Eurostat.

To compare redistribution, Table 2 reports the average Gini index between 1995 and 2007, for income before and after taxes. The data come from the OECD Income Inequality Database.¹⁴ Before-tax income Gini coefficients are similar across countries. For France, the higher pre-tax Gini reflects the treatment of public pensions as transfers rather than income, which mechanically raises pre-tax inequality. After-tax income Gini indices diverge: transfers in France reduce the Gini by about 20 points, compared with less than 10 points in the United States. France’s fiscal system is thus more redistributive than the US system.

The last column reports wealth Gini indices. For France, we use the 2010 Household Finance and Consumption Survey (HFCS), which is the closest wave to our benchmark window. The Gini index is stable across waves. For the United States, we use the PSID in 2006.¹⁵ Wealth inequality is substantially higher than post-tax income inequality in both countries (by more than 35 points), and remains about 10 points higher in the United States than in France.

¹³Extending the sample to 1995–2021 does not alter the main tax results, but the pre-2008 window is more consistent with the steady-state analysis.

¹⁴See <https://stats.oecd.org/index.aspx?queryid=66670>.

¹⁵The 2007 SCF yields a wealth Gini of 0.78, close to the PSID value.

Although Table 2 lists a *linear* labor tax, our model uses a *progressive* labor income tax consistent with both countries' systems. We estimate the HSV functional form for the labor tax:

$$\text{Tax:} \quad T(I_c) = I_c - \kappa I_c^{1-\tau}, \quad (36)$$

$$\text{Disposable income:} \quad D(I_c) = \kappa I_c^{1-\tau}, \quad (37)$$

where I_c denotes labor income in country c , τ is progressivity, and κ governs the level of taxation. This formulation enables cross-country comparisons despite a complex combination of heterogeneous schedules and multiple deductions in each country.

We estimate labor tax progressivity using the Luxembourg Income Study (LIS) database for France and the United States in 2005.¹⁶ Regressing the log of disposable income on the log of labor income (i.e., the log of (37)) yields the estimates in Table 3, where $\hat{\tau}$ is the estimated labor tax progressivity and SE is the associated standard error. France exhibits substantially greater progressivity. Our US estimate $\hat{\tau} = 0.16$ is close to values in the literature. It is slightly below Heathcote et al. (2017)'s value of 0.181, since we focus on labor income only instead of labor and capital income.

	$\hat{\tau}$	SE	Obs.	R^2
France	0.23	0.0056	5289	0.855
United States	0.16	0.0019	38111	0.942

Table 3: Estimates $\hat{\tau}$ of the progressivity of the labor income tax in the United States and in France for 2005.

We regress the log of equation (37) using the LIS database: SE is the standard error of $\hat{\tau}$, Obs. is the number of observations, R^2 is the R^2 of this regression.

5.2 Calibration of the model

We now calibrate the model to reproduce the inequality and fiscal aggregates described above. The model period is a quarter across all calibrations.

The US calibration

The calibration parameters are gathered in Table 4, and we detail below our calibration strategy.

¹⁶We restrict attention to heads of households and their spouses aged between 25 and 60 who were employed. We define *labor income* as the sum of wage income, self-employment income, and private transfers. Using Trabandt and Uhlig (2011) to infer capital tax, we compute the amount of capital income tax. We then subtract the capital tax amount from total LIS income taxes to estimate labor income tax. *Disposable income* is then *labor income* minus the *labor income tax*.

Preference parameters. Preferences are separable in consumption and labor. We set $\beta = 0.992$ to match a realistic capital-output ratio. The period utility function is $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ and $v(l) = \frac{1}{\chi} \frac{l^{1/\varphi+1}}{1/\varphi+1}$. The parameter γ governs the intertemporal elasticity of substitution and is set to $\gamma = 1.8$ to match wealth inequality at the targeted capital-output ratio. The Frisch elasticity of labor supply is set to $\varphi = 0.5$ following Chetty et al. (2011). We choose $\chi = 0.0477$ to normalize aggregate hours to 0.3.

Technology. The production function is $F(K, L, s) = sK^\alpha L^{1-\alpha} - \delta K$, with embedded depreciation. We set the capital share to $\alpha = 0.36$ and the quarterly depreciation rate to $\delta = 2.5\%$ (10% annualized).

Idiosyncratic labor risk. Productivity follows an AR(1) process: $\log y_t = \rho_y \log y_{t-1} + \varepsilon_t^y$ with $\varepsilon_t^y \sim \mathcal{N}(0, \sigma_y^2)$. We set $\rho_y = 0.99$ and $\sigma_y = 0.0995$, close to Krueger et al. (2018), and discretize the process with 10 states via Tauchen (1986). Together with the fiscal calibration below, this process helps reproduce a US post-tax income Gini index of 0.40, reported in Table 2.

Taxes and government budget constraint. We adopt the consumption and capital tax rates of Trabandt and Uhlig (2011): $\tau^K = 36\%$ and $\tau^c = 5\%$. Labor-tax progressivity is set to our estimate $\tau = 0.16$ from Table 3. We then choose the labor-tax level parameter $\kappa = 0.85$ to match a public-spending-to-GDP ratio of 15%. The resulting κ is close to the value in Ferriere and Navarro (2023) and the model-implied public-debt-to-GDP ratio matches the value of 63% reported in Table 2.

The calibrated US model replicates the main macro ratios and distributional targets. Consumption over GDP is 58% in the model, compared with 60% in the data for 1995-2007. The investment-to-GDP ratio is 27%, close to the empirical value of 25%. The post-tax income Gini is 0.40 in both the model and the data, while the wealth Gini index of 0.78 in the model is close to its empirical counterpart of 0.77. Table 5 summarizes these model-data comparisons. This fit indicates that the tax system provides a reasonable approximation to the income and wealth distribution in the United States, and hence of the redistributive effects of the US tax system. Similar results are found in Heathcote et al. (2017) and Dyrda and Pedroni (2022).

French calibration

The French calibration follows the same structure as the US calibration. The model period and functional forms are unchanged, while fiscal parameters, the productivity process, and selected targets are set to match French data. Table 4 reports the resulting parameter values.

France					
Parameter	Description	Value	Target or ref.	Value	Target or ref.
<i>Preference parameters</i>					
β	discount factor	0.992	$K/Y = 2.7$	0.996	$K/Y = 3.1$
u	utility function	.	$\gamma = 1.8$	.	$\gamma = 1.8$
φ	Frisch elasticity	0.5	Chetty et al. (2011)	0.5	Chetty et al. (2011)
χ	hours worked	0.33	Penn World Table	0.29	Penn World Table
α	capital share	36%	Profit Share, NIPA	36%	Profit Share, INSEE
δ	depreciation rate	2.5%	Chetty et al. (2011)	2.5%	Own calc., INSEE
<i>Productivity parameters</i>					
σ^y	std. err. prod.	0.10	Gini for income	0.06	Fonseca et al. (2023)
ρ^y	autocorr. prod.	0.99	Gini for income	0.99	Fonseca et al. (2023)

Table 4: Parameter values.

Preference parameters. We use the same preference specification as in the United States calibration. The discount factor and Frisch elasticity are set to $\beta = 0.996$ and $\varphi = 0.5$. We choose $\chi = 0.0228$ to normalize aggregate hours to 0.3, and again set the inverse of the intertemporal elasticity of substitution to $\gamma = 1.8$ to match inequality targets.

Technology and TFP shock. We keep the same production function as in the United States calibration: $F(K, L, s) = sK^\alpha L^{1-\alpha} - \delta K$ with $\alpha = 0.36$ and $\delta = 2.5\%$. This matches the targeted French aggregate ratios reported below.

Idiosyncratic risk. The AR(1) productivity process uses $\rho_y = 0.99$ and $\sigma_y = 0.0646$, which are consistent with Fonseca et al. (2023). As for the United States, we discretize with 10 states.

Taxes and government budget constraint. We set the fiscal instruments to match the French system reported in Table 2: a capital tax of $\tau^K = 35\%$, a progressivity parameter of $\tau = 0.23$, and a consumption tax of $\tau^c = 18\%$. We choose $\kappa = 0.728$ to match a public-spending-to-GDP ratio of 24%. The resulting public-debt-to-GDP ratio is 60%, as in Table 2.

The calibrated model matches French targets closely. Aggregate private consumption is 44% of GDP, close to the 45% estimated by Trabandt and Uhlig (2011) for 1995-2007. Investment amounts to 31% of GDP, equal to its empirical counterpart. The Gini index of 0.28 for post-tax income and of 0.68 for wealth also match the values of Table 2. Table 5 summarizes these model-data comparisons.

		France			
Parameter	Description	Model	Data	Model	Data
<i>Public finance aspects</i>					
B/Y	Public debt (%GDP)	63%	63%	60%	60%
G/Y	Public spending (%GDP)	15%	15%	25%	24%
<i>Aggregate quantities</i>					
C/Y	Aggregate consumption (%GDP)	58%	60%	44%	45%
I/Y	Aggregate investment (%GDP)	27%	25%	31%	31%
<i>Inequality measures</i>					
	Gini for post-tax income	40%	40%	28%	28%
	Gini for wealth	78%	77%	68%	68%

Table 5: Model implications for key variables.

Empirical values are discussed in Section 5.1 and summarized in Table 2.

5.3 Recovering G-SWF weights

We recover the current-productivity G-SWF weights by applying the inverse-optimal restrictions from Section 4.3 to the truncated Ramsey system derived in Appendix D. The truncation length is $N = 5$ in the baseline, so the approximation contains $10^5 = 100,000$ truncated histories. While histories are used to evaluate the Ramsey restrictions, the recovered welfare object remains the current-productivity G-SWF.

Baseline parametric weights. The baseline follows Definition 1 and imposes the three-parameter log-quadratic profile (35). This exactly matches the two independent Ramsey restrictions plus the normalization. Applying the parametric algorithm of Section 4.5 yields:

$$\begin{aligned}\log \omega(y)^{us} &= -0.25 + 1.06 \log(y) + 0.22(\log(y))^2, \\ \log \omega(y)^{fr} &= -0.51 + 0.62 \log(y) + 1.44(\log(y))^2.\end{aligned}$$

Figure 1 reports the resulting G-SWF weights.

The US profile rises with productivity. The French profile is U-shaped, with relatively high weights at both ends of the productivity distribution and lower weights in the middle. These profiles summarize the revealed marginal valuations that rationalize each country’s fiscal system. In the United States, the increasing profile reflects a system that relies less on redistribution across productivity states: high-productivity agents receive above-average recovered weights, while low-productivity agents receive weights close to zero. In France, the recovered weights are more compressed and non-monotone. The high weights at the bottom are consistent with stronger redistribution toward low-productivity households, while the high weights at the top reflect the role of capital taxation and saving behavior in the Ramsey restrictions. Thus, relative to the US, the French fiscal system

implies a more compressed pattern of revealed valuations, but not a monotone tilt toward low-productivity agents.

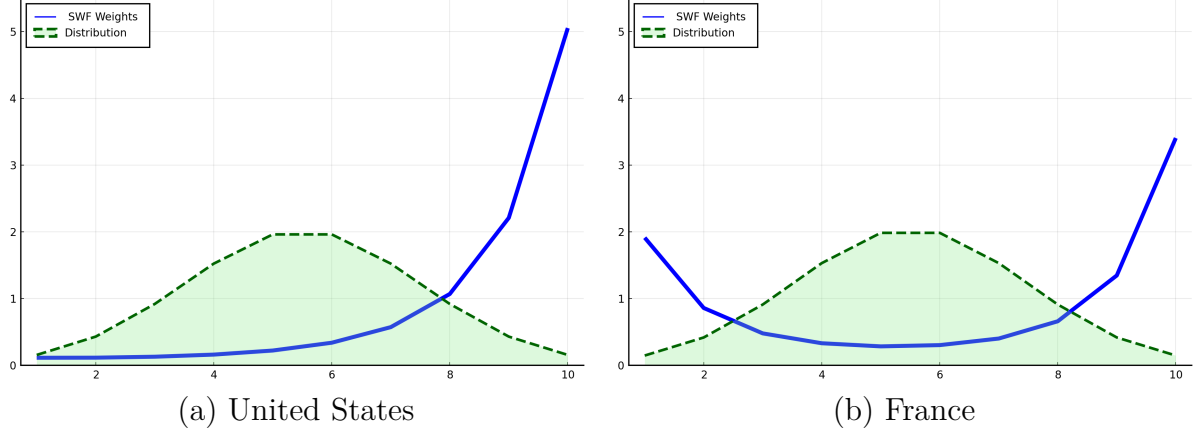


Figure 1: Parametric G-SWF weights as a function of productivity for the United States and France.

Non-parametric robustness. We then compute non-parametric G-SWF weights, defined as the minimum-variance weights satisfying the same Ramsey first-order conditions (see Appendix F.1 for the formal program). This selects, among the weights consistent with the observed fiscal system, the profile closest to the utilitarian benchmark. Figure 2 displays these weights against the income distribution, and Table 6 reports their distributional statistics.

In the United States, non-parametric G-SWF weights increase monotonically with productivity, consistent with the limited redistribution embedded in the US fiscal system. The dispersion is substantial: the bottom 10% of the productivity distribution receives a near-zero weight ($\omega = 0.006$), while the top 10% receives a weight of 2.96, a ratio of roughly 490 to 1. In France, the profile is non-monotone: weights are high at the bottom of the productivity distribution, fall sharply around the lower-middle states, and then rise again toward the top. The dispersion is markedly smaller than in the United States, with a standard deviation of 0.49 rather than 1.37. The two robust features are therefore compression of the French profile relative to the US profile and elevated recovered valuations for low-productivity agents in France.

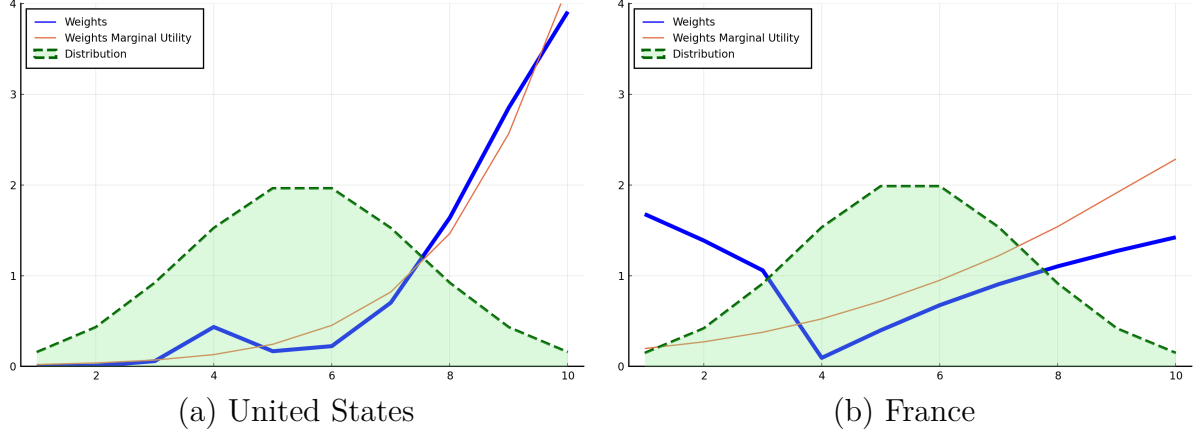


Figure 2: Non-parametric G-SWF weights ω_y (solid blue line), ex-post neutral benchmark $1/\bar{u}'(c_y)$ (solid red line), and income distribution (green shaded area). Both weight series are normalized by their respective means.

	United States		France	
	Non-param.	Parametric	Non-param.	Parametric
Mean	1.00	1.00	1.00	1.00
Standard deviation	1.37	1.57	0.49	1.00
Minimum	0.006	0.115	0.095	0.286
Maximum	3.91	5.05	1.68	3.41
Bottom 10%	0.006	0.117	0.37	0.303
Median	0.33	0.282	1.08	0.571
Top 10%	2.96	2.50	1.45	2.06

Table 6: Distributional statistics of non-parametric and parametric G-SWF weights for the US and France. Weights are normalized so that $\sum_y \pi_y \omega_y = 1$. Bottom 10%, Median, and Top 10% refer to weight values at the corresponding percentiles of the productivity distribution.

The redistributive spectrum. Figure 2 also displays the series $\bar{u}'(c_y)^{-1}$ (solid red line), the inverse of the average marginal utility of consumption at each productivity level, normalized by its mean. This series increases with productivity for a mechanical reason: higher-productivity agents consume more, and have lower marginal utility. It helps translate recovered G-SWF weights ω_y into Social Marginal Welfare Weights (SMWW) $\lambda_y = \omega_y \bar{u}'(c_y)$. If ω_y were proportional to $\bar{u}'(c_y)^{-1}$, the implied SMWW would be constant across productivity levels, meaning that the planner places equal social value on a marginal unit of resources regardless of who receives it. Under a utilitarian planner, ω_y is constant, so the SMWW is proportional to $\bar{u}'(c_y)$ and decreases with y , reflecting a preference for redistribution toward low-productivity agents. Appendix F.3 constructs λ_y in our calibrated framework and shows that it aligns closely with existing microeconomic evidence.

Parametric and non-parametric comparison. The log-quadratic baseline and the minimum-variance profile deliver the same qualitative country patterns. In the United States, both approaches recover a profile tilted toward high-productivity states. In France, both recover a more compressed and non-monotone profile, with elevated weights at the bottom and a smaller rise at the top. The non-parametric exercise therefore shows that the qualitative country patterns are not driven mechanically by the log-quadratic functional form. A more detailed comparison is provided in Appendix F.1.

5.4 Diagnostic decomposition of cross-country weight differences

We next run diagnostic decompositions that replace, one at a time or jointly, the calibrated ingredients that differ across the two economies: the discount factor, the tax system, and the productivity process. After each replacement, we simulate the model and recover the G-SWF with the same inverse-optimal approach. The exercise is descriptive: it shows how recovered G-SWF weights move within the model when specific ingredients are changed.

Discount factor. Replacing only the discount factor moves the recovered profiles in the expected direction but leaves the main country patterns intact. In the US calibration, France’s higher discount factor raises weights at the bottom but preserves the overall top-weighted shape (see the red dotted line in panel (a) of Figure 3). In the French calibration, the US discount factor lowers bottom weights and raises top weights (see the red dotted line in panel (b) of Figure 3). The discount-factor replacement therefore moves the recovered profile in the expected direction, but does not by itself explain the US-France difference.

Tax system. Replacing the tax system has the largest effect on recovered weights. In the US calibration, imposing the French tax system shifts recovered G-SWF weights away from high-productivity states and toward the bottom of the distribution. See the orange dashed line in panel (a) of Figure 3 for the US weights under the French tax system and French discount factor β . Conversely, imposing the US tax system in the French calibration raises top weights and lowers bottom weights. See panel (b) of Figure 3 for the French weights with the US tax system and discount factor. The fiscal system is therefore the main diagnostic channel: higher labor-tax progressivity implies larger recovered G-SWF weights for low-productivity agents.

Productivity process. Finally, we replace the United States income process with France’s, together with the French discount factor and taxes. The resulting G-SWF weights reproduce the French weights (blue line in panel (a) of Figure 3). The same holds for France in panel (b). This final replacement is a consistency check: once all calibrated

ingredients are jointly replaced, the recovered profile coincides with that of the other country.

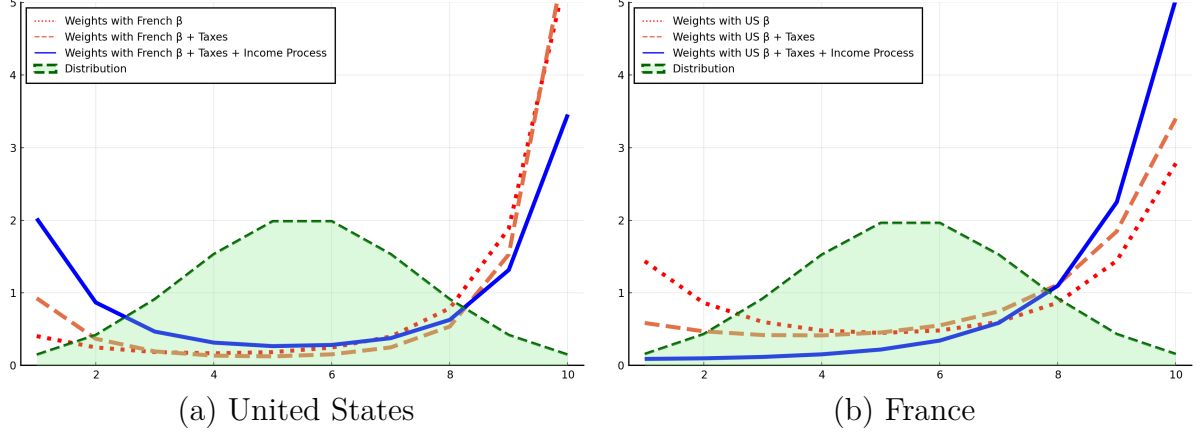


Figure 3: Diagnostic decompositions of US and French G-SWF weights, varying discount factors, tax systems, and income processes.

Appendix E provides a more detailed accounting of the tax-system replacement. It shows that adopting the French tax system in the United States reduces labor income for high-productivity agents through lower labor-supply incentives, while increasing consumption and utility for low-productivity agents. This decomposition explains why higher labor-tax progressivity maps into larger recovered G-SWF weights at the bottom and smaller recovered weights at the top.

5.5 A French G-SWF illustration for the United States

We close the quantitative section with a within-model accounting exercise. Holding US preferences, technology, and the productivity process fixed, we search for fiscal instruments under which the inverse-optimal recovery rationalizes a G-SWF profile close to the French one. Transition dynamics remain outside the scope of this accounting exercise.

We proceed as follows. Consider a “core” US economy with US individual preferences, technology, and productivity process. We jointly choose the capital tax and the labor-tax progressivity to minimize the Euclidean distance between the resulting G-SWF weights and the French G-SWF weights. Because the two profiles are defined on different productivity grids, both are interpolated onto a common grid before computing the distance. After choosing these two instruments, we adjust the parameter κ to keep the public-spending-to-output ratio equal to the United States benchmark. Hence, at the solution, the main macro ratios (capital-to-output, investment-to-output, aggregate consumption to output, and public spending to output) match US values by construction.

The exercise produces a fiscal system and recovered G-SWF weights associated with that fiscal system. The recovered weights closely match the French profile, as shown in Figure 4.

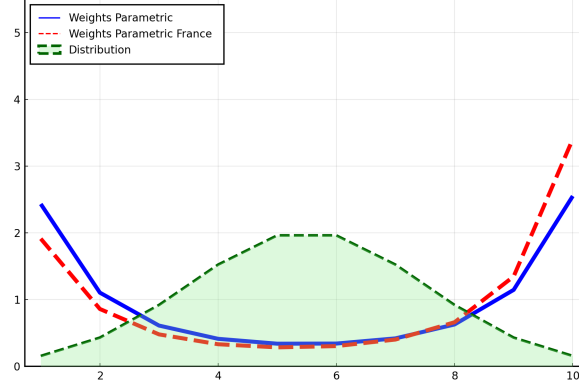


Figure 4: G-SWF weights for France (red dashed line) and for the US fiscal system that minimizes the distance to the French weights (blue line). The x-axis corresponds to the 10 productivity levels in France.

Table 7 reports the fiscal system produced by this exercise, alongside the baseline US and French systems. The associated inequality statistics are close to France’s and substantially lower than in the United States benchmark. The large implied public-debt ratio should therefore be viewed as an accounting outcome of the maintained constraints.

	Public debt (%GDP)	τ_k (%)	τ (%)	κ (%)	Gini post-tax income	Gini wealth
France	63	36	16	85	40	78
with French G-SWF	60	35	23	73	28	68
	299	9	57	71	27	63

Table 7: Comparison between the benchmark economies and the US economy with the French G-SWF profile.

Within-model accounting. Within this accounting exercise, the decline in inequality is mainly driven by the much higher labor-tax progressivity (from 16% to 57%), which mechanically redistributes toward the bottom. The remaining “core” parameters and the main macro ratios (C/Y , G/Y , I/Y , and, by construction, K/Y) are held at their US benchmark values by adjusting κ , while the pre-tax interest rate remains pinned down by $1/\beta$.

Higher progressivity reduces labor supply at the top by lowering the net-of-tax wage, which compresses the earnings dispersion. With the US productivity process, lower effective hours for high-productivity agents reduce efficient labor input. With K/L anchored by the pre-tax interest rate, the implied capital stock adjusts to keep the steady-state factor-price conditions. Because the model keeps aggregate ratios and productivity at US levels, saving demand is higher than in France. The model adjusts through public debt, which expands from 63% to 299% of GDP.

The capital-tax response partly offsets the increase in labor-tax progressivity for high-saving, high-productivity households. Lowering the capital tax raises the return to saving and helps the recovered profile match the upper tail of the French weights. It also raises saving demand, reinforcing the public-debt residual. The exercise also explains the low middle-productivity weights in the French profile: those agents bear part of the higher labor-tax progressivity but benefit less from the lower capital tax than high-saving households.

6 Maintained assumptions and robustness

This section clarifies the assumptions needed to interpret the recovered G-SWF weights. We first discuss the additive aggregation and cardinal-utility structure shared with utilitarian and weighted-utilitarian planners. We then report robustness to preference parameters under recalibrated steady states. Finally, we explain how history-dependence checks address the underidentification of G-SWF weights under the Ramsey restrictions.

6.1 Additive aggregation and cardinal utility

Linear aggregation assumption. We consider linear aggregation of history-based evaluations at the planner level. This assumption yields a transparent and tractable reduced-form H-SWF and keeps the inverse-optimal approach interpretable as a recovery of marginal social valuations. Linear expected-welfare criteria are standard in incomplete-insurance models. Aiyagari and McGrattan (1998), for example, use the utilitarian benchmark to evaluate the steady-state welfare. Our construction generalizes this logic by allowing the weights to depend on histories, and later on current productivity states. This is the margin along which our criterion departs from the utilitarian benchmark.

Departing from linear aggregation would introduce additional curvature at the social-criterion level that is difficult to separately identify empirically. As in the generalized social marginal welfare weight approach of Saez and Stantcheva (2016), the linear G-SWF can be understood as a first-order Taylor approximation to a more general social objective. This is also the approach followed in Olivi et al. (2025). The recovered G-SWF weights should therefore be read as local marginal social valuations around the observed allocation.

The cardinality of the utility function and estimated weights. Once a common utility specification is fixed, common positive affine normalizations of the period utility function only relabel the utility scale and leave normalized recovered weights unchanged. Arbitrary increasing transformations are different. Applied to period utility, or to the induced intertemporal utility, they generally change risk attitudes, intertemporal trade-offs, and the marginal-utility ratios used in the inverse-optimal approach.

However, this cardinality restriction is not specific to the G-SWF. It is also shared by the class of utilitarian and weighted-utilitarian criteria. The paper thus follows the standard additive social-welfare and inverse-optimal approach by taking a cardinal utility specification as part of the model. We assess sensitivity by varying preference parameters, including the intertemporal elasticity of substitution and the Frisch elasticity. We find that the recovered qualitative country patterns remain stable across these specifications. These results show that the French low-productivity pattern and the US high-productivity pattern hold across several utility parameterizations.

6.2 Robustness to preference parameters

A natural concern is whether the shape of the recovered G-SWF profiles is driven by the preference parameters used in the baseline calibration rather than by the structure of the observed fiscal system.¹⁷ We examine this concern by recalibrating the model under alternative values of the intertemporal elasticity of substitution and the Frisch elasticity.

For the IES exercise, we set $\sigma \in \{1, 2, 2.5\}$. For each value, we recalibrate the income-process parameters and fiscal instruments against the same targets as in the baseline: public debt, the wealth Gini, public spending, consumption, and investment. Varying the income process alongside the fiscal instruments is necessary because, away from the baseline curvature, the fiscal instruments alone cannot match both the observed wealth-to-output ratio and the wealth Gini. Appendix G reports the calibrated parameters and achieved moments. The flow ratios (G/Y , C/Y , and I/Y) and the Gini ratios are closely matched in both countries. Public debt absorbs the remaining calibration tension in some cases, especially for France.

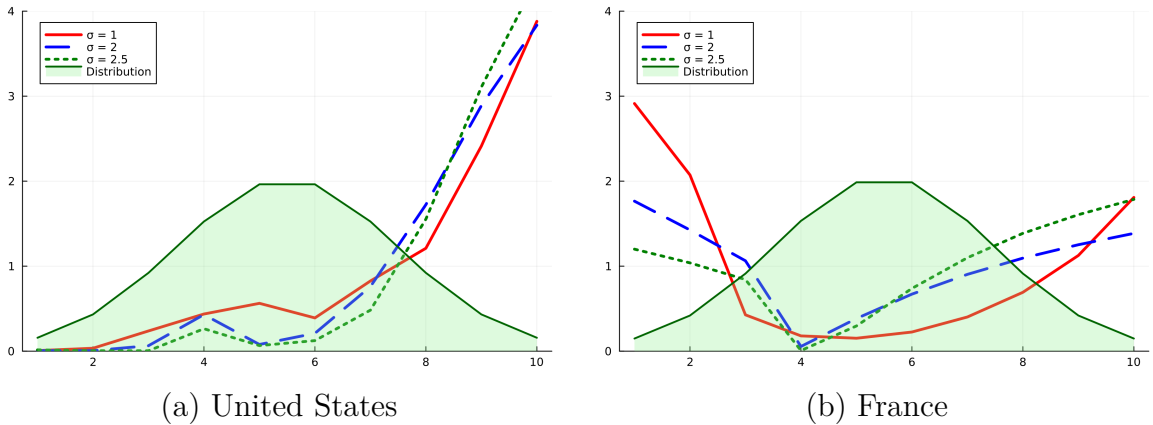


Figure 5: Non-parametric G-SWF weights ω_y for the inverse of IES $\sigma \in \{1, 2, 2.5\}$ and income distribution (green shaded area). The x-axis indexes the 10 productivity levels in ascending order.

¹⁷In quantitative heterogeneous-agent models, preference parameters are disciplined by calibration targets rather than chosen freely. The exercises below should therefore be read as sensitivity checks.

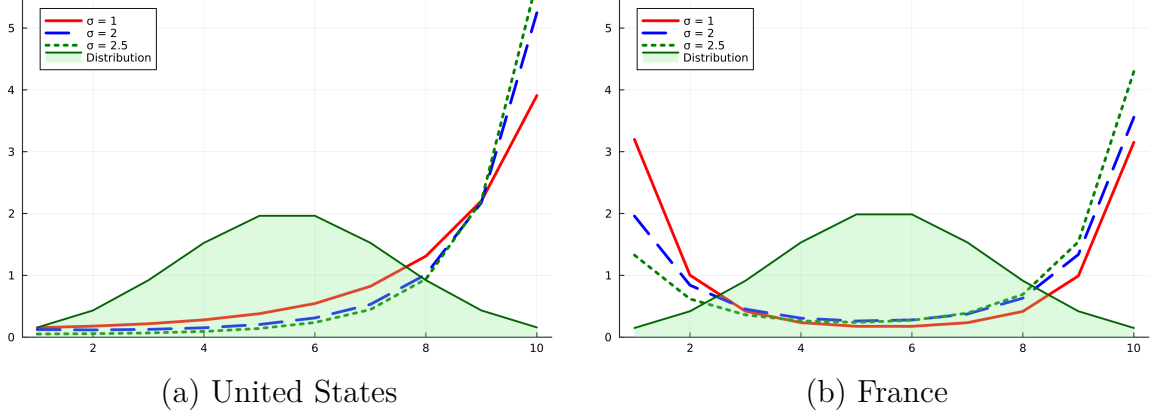


Figure 6: Parametric G-SWF weights ω_y for the inverse of IES $\sigma \in \{1, 2, 2.5\}$ and income distribution (green shaded area).

The recovered profiles are stable across the IES values. In the United States, both the parametric and non-parametric profiles remain broadly increasing across the productivity distribution for all three values of the inverse IES, closely reproducing the monotone pattern of the baseline (Figures 2 and 1). The weight on the least productive agents remains low, while the upper tail receives above-average weights. The slope of the profile varies with σ : under log utility ($\sigma = 1$) the increasing slope is steeper, reflecting that the absence of a precautionary-savings motive implies a stronger income effect, which pushes the distribution of weights toward the most productive agents; as σ rises toward 2.5, the profile flattens slightly as stronger precautionary savings provide more self-insurance and somewhat attenuate the dispersion of welfare across productivity levels. Crucially, however, the direction of the profile is unchanged: the US fiscal system consistently reveals larger valuations for high-productivity agents, irrespective of the assumed curvature.

For France, the profile remains non-monotone and U-shaped across the IES values. Weights are elevated at the bottom of the productivity distribution, fall in the middle, and rise again toward the top. The height of the tails changes somewhat with curvature, but the qualitative shape is preserved. The parametric profiles track the non-parametric profiles closely, indicating that the log-quadratic specification captures the main country-level patterns rather than imposing them mechanically.

Two observations reinforce this interpretation. First, the parametric profiles in Figure 6 closely track the non-parametric ones in Figure 5 for all values of σ . The log-quadratic specification therefore appears to correctly capture the main features of the weight profiles. Second, the income process is recalibrated across the exercises. For example, in the United States under $\sigma = 1$, σ^y rises from 0.196 to 0.23 to compensate for the absence of precautionary savings. The stability of the weight profiles despite this joint recalibration of income risk and fiscal instruments suggests that the revealed social valuations recovered in Section 5.3 are stable features of the observed tax-and-transfer system and not artifacts of the specific assumption on utility curvature. Appendix G compares each alternative- σ

profile with the baseline $\sigma = 1.8$ calibration.

We next vary the Frisch elasticity of labor supply, φ . The baseline calibration sets $\varphi = 0.5$ following Chetty et al. (2011). We repeat the exercise for $\varphi \in \{0.25, 0.75, 1.00, 1.25, 1.50\}$. For each value of the Frisch elasticity, we keep the structural parameters and income process at their respective country baseline values and recalibrate the fiscal parameters (τ, κ, t_c) against the same target set: B/Y , wealth Gini, G/Y , C/Y , and I/Y . The full calibrated parameters and achieved moments are reported in Appendix H. As in the IES exercise, the flow ratios are matched closely while public debt remains a residual margin in some calibrations, especially for France.

For both countries, the recovered weight profiles are stable across Frisch elasticities. In the United States, the profile remains increasing in productivity across the entire range of the Frisch elasticity and the profiles are nearly indistinguishable. In France, the non-monotone U-shaped profile is preserved. The progressivity parameter τ moves little and remains between 0.23 and 0.24 because the high level of government spending ($G/Y = 0.25$) leaves limited room for adjustment.

Taken together, the IES and Frisch exercises show that the recovered country weight profiles are stable across the preference specifications we consider here.

6.3 History-dependence checks

The baseline recovery via the inverse-optimal approach imposes current-productivity sufficiency: all agents with the same current productivity receive the same G-SWF weight. This restriction is operational, because current productivity is observed in the stationary cross section, but it may hide welfare-relevant differences across histories. We therefore ask whether allowing weights to depend on truncated histories changes the information recovered in the inverse-optimal approach.

Relaxing this current-productivity sufficiency assumption requires assigning separate weights to each truncated history, so the number of weights rises from Y (the number of productivity levels) to Y^N . The Ramsey restrictions are then even more underidentified. We therefore recover history-dependent weights using the same minimum-variance criterion as in the non-parametric exercise. To compare the result with the baseline G-SWF, we average the recovered history weights within each current-productivity state, weighting histories by their stationary masses. The averaged history-dependent profile is close to the non-parametric current-productivity profile. This suggests that the baseline G-SWF profile for both countries is not an artifact of imposing current-productivity sufficiency. Further details are provided in Appendix F.2.

Overall, the recovered country patterns are stable along the main margins that matter for interpretation. They are robust to changes in preference parameters, the replacement of the log-quadratic profile by a minimum-variance non-parametric recovery, and the

relaxation of current-productivity sufficiency toward truncated-history weights.

7 Conclusion

We develop a history-based welfare framework for heterogeneous-agent economies and study a stationary restriction, the G-SWF, whose weights depend on current observable productivity levels. This restriction allows the planner to depart from the utilitarian benchmark while remaining recoverable from stationary data. We then show how to recover G-SWF weights from observed fiscal systems and distributional outcomes using an inverse-optimal Ramsey approach.

Applying the method to France and the United States, we find markedly different recovered welfare profiles. The French profile assigns more weight to low-productivity agents and is more compressed overall. The US profile rises with productivity, assigning larger recovered weights to high-productivity agents. These differences rationalize distinct fiscal systems within the maintained Ramsey model and illustrate how revealed social valuations can vary across countries. Under our recovery exercise, the United States appears closer to a revealed-valuation profile associated with less redistribution, while France appears closer to an egalitarian one, particularly at lower productivity levels.

These results open new avenues for research, especially regarding the stability of recovered social valuations over time. A key next step is to examine the role of the G-SWF in shaping fiscal responses to economic shocks, particularly in business cycle stabilization.

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Online Appendix

A Proofs related to the G-SWF construction

A.1 Proof of Proposition 1

By history-realization consequentialism, the planner's ranking over allocations can be represented by a continuous function of the finite vector of history-realization utilities, whose components are $u(c_{1,y})$ for period-1 histories and $u(c_{2,y,y'})$ for period-2 histories. The separability condition gives an additive representation: up to an increasing transformation, the criterion can be written as a sum of component functions, one for each history realization. The constant-social-value condition then restricts each component to be affine in its own utility argument. Hence the criterion is represented, up to an irrelevant additive constant and a positive affine normalization, by a weighted sum of history-realization utilities. Monotonicity then implies that the weights are nonnegative.

Hence, up to an increasing affine transformation, the planner's criterion can be written as a weighted sum of the history-realization utilities:

$$\sum_{y \in \mathcal{Y}} \alpha_{1,y} u(c_{1,y}) + \sum_{y,y' \in \mathcal{Y}} \alpha_{2,y,y'} u(c_{2,y,y'}).$$

Monotonicity implies $\alpha_{1,y} \geq 0$ and $\alpha_{2,y,y'} \geq 0$.

For positive-mass history realizations, define $\omega_{1,y} := \alpha_{1,y}/\pi_y$ and $\omega_{2,y,y'} := \alpha_{2,y,y'}/(\beta\pi_{yy'})$. If a history realization has zero probability, its utility component receives zero population mass in the represented criterion, so its associated weight is immaterial and can be assigned any nonnegative value. Substituting these definitions on positive-mass realizations gives:

$$\mathcal{W}^H = \sum_{y \in \mathcal{Y}} \pi_y \omega_{1,y} u(c_{1,y}) + \beta \sum_{y,y' \in \mathcal{Y}} \omega_{2,y,y'} \pi_{yy'} u(c_{2,y,y'}),$$

up to the same positive affine normalization. This is the claimed H-SWF representation.

A.2 Proof of Proposition 4

Let:

$$\nu_0^\omega(B) := \int_{\mathcal{I}} \omega_i \mathbf{1}\{(a_{i,0}, y_{i,0}) \in B\} di$$

for Borel sets $B \subseteq \mathcal{A} \times \mathcal{Y}$. Because $\omega_i \geq 0$ and $\int_{\mathcal{I}} \omega_i di = 1$, ν_0^ω is a probability measure. Let ν_t^ω be the corresponding Pareto-weighted distribution of current states at date t induced by the stationary policy functions and Π .

The PN-SWF can therefore be written as:

$$W_t^P = \int_{\mathcal{A} \times \mathcal{Y}} \mathcal{V}(a, y) \nu_t^\omega(da, dy).$$

By common ergodicity, the Pareto-weighted distribution ν_t^ω converges weakly to the invariant distribution μ^* . In the finite-grid implementation, $\mathcal{V}$ is bounded, so this convergence implies:

$$\lim_{t \rightarrow \infty} W_t^P = \int_{\mathcal{A} \times \mathcal{Y}} \mathcal{V}(a, y) \mu^*(da, dy).$$

In the general case, we impose the corresponding integral-convergence condition through uniform integrability of $|\mathcal{V}|$ under the $(\nu_t^\omega)_t$.

It remains to evaluate the last integral. Using the invariance of the stationary distribution, we obtain:

$$\begin{aligned} \int_{\mathcal{A} \times \mathcal{Y}} \mathcal{V}(a, y) \mu^*(da, dy) &= \sum_{s=0}^{\infty} \beta^s \int_{\mathcal{A} \times \mathcal{Y}} \mathbb{E}_{a,y} [u(c(a_s, y_s))] \mu^*(da, dy) \\ &= \sum_{s=0}^{\infty} \beta^s \int_{\mathcal{A} \times \mathcal{Y}} u(c(a, y)) \mu^*(da, dy) \\ &= \frac{1}{1 - \beta} \int_{\mathcal{A} \times \mathcal{Y}} u(c(a, y)) \mu^*(da, dy). \end{aligned}$$

This expression does not depend on (ω_i) , which proves the proposition.

A.3 Construction of the measure on the set of idiosyncratic histories $\mathcal{Y}^\infty$

We can construct a probability space related to the set of infinite idiosyncratic histories, $\mathcal{Y}^\infty$. We summarize here the construction, and further details can be found in LeGrand and Ragot (2022a, Appendix B.3). Consistent with the main text, we will typically denote by y^t an element of $\mathcal{Y}^\infty$. Such an element $\tilde{y}^t \in \mathcal{Y}^\infty$ can be described as a left-infinite sequence:

$$\tilde{y}^t = (\dots, y_{-k}(\tilde{y}^t), \dots, y_{-1}(\tilde{y}^t), y_0(\tilde{y}^t)),$$

where each $y_{-k} : \mathcal{Y}^\infty \rightarrow \mathcal{Y}$ is a coordinate function returning the idiosyncratic state k periods ago. For the sake of simplicity, and as in the main text, we will denote by $y_s^t := y_{-(t-s)}(y^t)$ for any $s \leq t$ the state at date s , which is $t - s$ periods ahead of date t .

We define $L(y^t)$ as the past of history y^t – which discards the current state y_t^t :

$$L(y^t) = (\dots, y_{t-k}^t, \dots, y_{t-1}^t).$$

Consistent with the main text, we also denote by $y^{t-1,t}$ the past history of y^t : $y^{t-1,t} = L(y^t)$.

We can then define the cylinder sets $C_k(A)$ for any $k \geq 1$ and any $A \subset \mathcal{Y}^k$ as follows:

$$C_k(A) = \{y^t \in \mathcal{Y}^\infty : (y_{t-k+1}^t, \dots, y_t^t) \in A\}.$$

The cylinder set $C_k(A)$ is the subset of $\mathcal{Y}^\infty$ containing all idiosyncratic histories whose truncation of length k belongs to A . We then define $\mathcal{C}_0$ as the set of all cylinder sets, which can be shown to be a field (Billingsley, 2012, Section 2). We denote by $\mathcal{F} := \sigma(\mathcal{C}_0)$ the cylindrical σ -algebra generated by $\mathcal{C}_0$, and we define the set function $\mu : \mathcal{C}_0 \rightarrow \mathbb{R}$ from the transition matrix Π and the stationary vector π , such that for any $k \geq 2$ and any $A \subset \mathcal{Y}^k$:

$$\begin{cases} \mu(C_k(A)) = \sum_{(y_{-k+1}, \dots, y_0) \in A} \pi_{y_{-k+1}} \Pi_{y_{-k+1} y_{-k+2}} \dots \Pi_{y_{-1} y_0} & \text{for any } k \geq 2 \text{ and } A \subset \mathcal{Y}^k, \\ \mu(C_1(A)) = \sum_{y_0 \in A} \pi_{y_0} & \text{for any } A \subset \mathcal{Y}. \end{cases} \quad (38)$$

Finally, we can state the following lemma.

Lemma 3. *The triplet $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$ is a probability space.*

Proof. A proof can be found in Billingsley (2012, Section 2). The key part of the proof is to extend the measure μ defined on $\mathcal{C}_0$ to a measure defined on $\sigma(\mathcal{C}_0) = \mathcal{F}$. A consequence of Lemma 3 is that $\mu(\mathcal{Y}^\infty) = 1$, or $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$. $\square$

A.4 The conditional measure

To lighten formulas, for any $(y_{-k+1}, \dots, y_0) \in \mathcal{Y}^k$, we define $y_{-k+1:0} := (y_{-k+1}, \dots, y_0)$ as the vector of length k , containing elements whose indices range from $-k+1$ to 0. Similarly, for any $\tilde{y}^t \in \mathcal{Y}^\infty$, $\tilde{y}_{(t-k+1:t)}^t = (\tilde{y}_{t-k+1}^t, \dots, \tilde{y}_t^t)$ is the vector of the last k realizations of $\tilde{y}^t$. We define μ_1 by:

$$\mu_1(C_1(Y_1)|\tilde{y}^t) = \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}}, \text{ for any } Y_1 \subset \mathcal{Y}, \quad (39)$$

$$\forall k \geq 2, \mu_1(C_k(Y_k)|\tilde{y}^t) = \sum_{y_{(t-k+2:t+1)} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{t-k+2:t}^t}, \text{ for any } Y_k \subset \mathcal{Y}^k, \quad (40)$$

where for any elements $x, \tilde{x}$ of the same set, $1_{x=\tilde{x}} = 1$ if $x = \tilde{x}$ and $1_{x=\tilde{x}} = 0$ otherwise. Intuitively, the expression in (40) sums, over all possible vectors $y_{(t-k+2):t+1}$ of length k , the probability of switching from $\tilde{y}^t$ to a history ending in $y_{(t-k+2):t+1}$. The latter probability is equal to the probability of switching from $\tilde{y}_t^t$ to y_{t+1} , provided that $\tilde{y}^t$ and $y_{(t-k+2):t}$ are compatible (i.e., the $k-1$ last realizations of $\tilde{y}^t$ equal $y_{(t-k+2):t}$). Note that we could also write: $1_{y_{(t-k+2):t} = \tilde{y}_{t-k+2:t}^t} = \prod_{j=0}^{k-2} 1_{y_{t-j} = \tilde{y}_{t-j}^t}$.

Lemma 4. *For any $\tilde{y}^t \in \mathcal{Y}^\infty$, the set function $C \in \mathcal{C}_0 \mapsto \mu_1(C|\tilde{y}^t)$ is a pre-measure.*

Proof. In the remainder of the proof, we set $\tilde{y}^t \in \mathcal{Y}^\infty$. As a preliminary, we state two properties that we will use extensively below:

– For all $k' \geq k \geq 1$, for all $Y_k \subset \mathcal{Y}^k$, for all $Y'_{k'-k} \subset \mathcal{Y}^{k'-k}$:

$$\begin{aligned} & \sum_{y_{(t-k'+2):t+1} \in Y'_{k'-k} \times Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t} = \\ & \sum_{y_{(t-k+2):t+1} \in Y_k} \left(\Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} \times \right. \\ & \left. \sum_{y_{(t-k'+2):(t-k+1)} \in Y'_{k'-k}} 1_{y_{(t-k'+2):(t-k+1)} = \tilde{y}_{(t-k'+2):(t-k+1)}^t} \right). \end{aligned} \quad (41)$$

– For all $k' \geq k \geq 0$:

$$\sum_{y_{t-k':t-k} \in \mathcal{Y}^{k'-k+1}} 1_{y_{t-k':t-k} = \tilde{y}_{t-k':t-k}^t} = 1. \quad (42)$$

In the remainder, these will be referred to by their equation numbering. The proof of (42) is straightforward and comes from the fact that $(\tilde{y}_{t-k'}^t, \dots, \tilde{y}_{t-k}^t)$ is a unique element of the set $\mathcal{Y}^{k'-k+1}$. For the proof of (41), we denote by $S_{k,k'}$ the left-hand side. We have:

$$\begin{aligned} S_{k,k'} = & \sum_{y_{(t-k+2):t+1} \in Y_k} \sum_{y_{(t-k'+2):(t-k+1)} \in Y'_{k'-k}} \left(\Pi_{\tilde{y}_t^t y_{t+1}} \times \right. \\ & \left. 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} 1_{y_{(t-k'+2):(t-k+1)} = \tilde{y}_{(t-k'+2):(t-k+1)}^t} \right), \end{aligned} \quad (43)$$

where we have used the properties of a sum on a product space and the fact that $(x, y) = (x', y')$ iff $x = x'$ and $y = y'$ (where (x, x') and (y, y') are two pairs of vectors of the same length). Then we can factorize $\Pi_{\tilde{y}_t^t y_{t+1}}$ and $1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t}$ in (43), as they are independent from the sum over $y_{(t-k'+2):(t-k+1)} \in Y'_{k'-k}$. We then readily obtain (41).

We now go back to the proof of Lemma 4. Three properties need to hold: (i) well-defined, (ii) (countably) additive, (iii) $\mu_1(\mathcal{Y}^\infty | \tilde{y}^t) = 1$ for all $\tilde{y}^t \in \mathcal{Y}^\infty$.

For point (i), we need to check that $\mu_1(C_k(Y_k) | \tilde{y}^t) = \mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k) | \tilde{y}^t)$ for all $k' \geq k$. We have for $k \geq 2$:

$$\begin{aligned} \mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k) | \tilde{y}^t) &= \sum_{y_{(t-k'+2):t+1} \in \mathcal{Y}^{k'-k} \times Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t}, \\ &= \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} = \mu_1(C_k(Y_k) | \tilde{y}^t), \end{aligned} \quad (44)$$

where the first and last equalities use the definition of μ_1 , and the second the combination of properties (41) and (42).

We need to prove the result for $k = 1$. We have for $k' \geq 1$:

$$\begin{aligned}\mu_1(C_{k'}(\mathcal{Y}^{k'-1} \times Y_1)|\tilde{y}^t) &= \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} \left(\sum_{y_{(t-k'+2):t} \in \mathcal{Y}^{k'-1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t} \right), \\ &= \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} = \mu_1(C_1(Y_1)|\tilde{y}^t),\end{aligned}$$

where the first equality combines the definition of μ_1 and (41), the second uses the property (42), and the last the definition of μ_1 .

For point (ii), we consider two disjoint cylinders, $C_k(Y_k)$ and $C_{k'}(Y'_{k'})$ ($k' \geq k$, $Y_k \subset \mathcal{Y}^k$ and $Y'_{k'} \subset \mathcal{Y}^{k'}$). Since both cylinders are disjoint, then $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint too. We deduce that:

$$\begin{aligned}\mu_1(C_k(Y_k) \cup C_{k'}(Y'_{k'})|\tilde{y}^t) &= \mu_1(C_{k'}((\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'})|\tilde{y}^t), \\ &= \sum_{y_{(t-k'+2):t+1} \in (\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'}} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t}, \\ &= \sum_{y_{(t-k'+2):t+1} \in (\mathcal{Y}^{k'-k} \times Y_k)} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t} \\ &\quad + \sum_{y_{(t-k'+2):t+1} \in Y'_{k'}} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t}, \\ &= \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} \\ &\quad + \sum_{y_{(t-k'+2):t+1} \in Y'_{k'}} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}_{(t-k'+2):t}^t}, \\ &= \mu_1(C_k(Y_k)|\tilde{y}^t) + \mu_1(C_{k'}(Y'_{k'})|\tilde{y}^t),\end{aligned}$$

where the first equality uses the algebra property of cylinder sets, the second the definition of μ_1 , the third the property that $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint, the fourth the combination of properties (41) and (42), and the last the definition of μ_1 twice. We have thus proved that $\mu_1(\cdot|\tilde{y}^t)$ is finitely additive.

For point (iii), let $k \geq 2$. We have:

$$\begin{aligned}\mu_1(C_k(\mathcal{Y}^k)|\tilde{y}^t) &= \sum_{y_{t+1} \in \mathcal{Y}} \Pi_{\tilde{y}_t^t y_{t+1}} \sum_{y_{(t-k+2):t} \in \mathcal{Y}^{k-1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t}, \\ &= \sum_{y_{t+1} \in \mathcal{Y}} \Pi_{\tilde{y}_t^t y_{t+1}} = 1,\end{aligned}$$

where the first equality uses the definition of μ_1 and (41), the second property (42), and the third the property of the transition matrix Π . We thus deduce that $\mu_1(\mathcal{Y}^\infty|\tilde{y}^t) = 1$.

We have proven that $\mu_1(\cdot|\tilde{y}^t)$ is a finitely additive probability measure on the algebra $\mathcal{C}_0$. Billingsley (2012, Theorem 2.3) states that any finitely additive probability measure on the cylinder algebra is countably additive. We thus conclude that μ_1 is a countably

additive probability measure on $\mathcal{C}_0$ and is therefore a pre-measure on $\mathcal{C}_0$. $\square$

We then prove the following lemma. We recall that $\mathcal{F}$ is the cylindrical σ -algebra, $\sigma(\mathcal{C}_0)$, generated by $\mathcal{C}_0$.

Lemma 5. *For all $\tilde{y}^t \in \mathcal{Y}^\infty$, the function $\mu_1(\cdot|\tilde{y}^t)$ uniquely extends to a measure on $\mathcal{F}$. The proof is similar to the one showing the extension of μ as a pre-measure on $\mathcal{C}_0$ to a measure on $\mathcal{F}$. It relies on the Hahn-Kolmogorov theorem (Billingsley 2012, Theorem 3.1). See LeGrand and Ragot (2022a, Lemma 3 in Section B.3).*

Finally, we can state the following lemma, showing that μ_1 is a conditional measure.

Lemma 6. *For all $\tilde{y}^t \in \mathcal{Y}^\infty$ and for all $F \in \mathcal{F}$:*

$$\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(F|\tilde{y}^t) \mu(d\tilde{y}^t) = \mu(F). \quad (45)$$

Proof. We first prove (45) for F being a cylinder set. $\square$

First, let $Y_1 \subset \mathcal{Y}$ and consider $C_1(Y_1)$. We have:

$$\begin{aligned} \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(C_1(Y_1)|\tilde{y}^t) \mu(d\tilde{y}^t) &= \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} \mu(d\tilde{y}^t), \\ &= \sum_{\tilde{y}_t^t \in \mathcal{Y}} \pi_{\tilde{y}_t^t} \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}}, \\ &= \sum_{y_{t+1} \in Y_1} \sum_{\tilde{y}_t^t \in \mathcal{Y}} \pi_{\tilde{y}_t^t} \Pi_{\tilde{y}_t^t y_{t+1}}, \\ &= \sum_{y_{t+1} \in Y_1} \pi_{y_0} = \mu(C_1(Y_1)), \end{aligned}$$

where the first equality comes from the definition (39) of μ_1 , the second from the fact that the integral is actually carried over a cylinder set of the form $C_1(\mathcal{Y})$ and from the definition (38) of μ , the third from the permutation of finite sums, the fourth from the fact that π is stationary ($\sum_{y \in \mathcal{Y}} \pi_y \Pi_{yy'} = \pi_{y'}$), and the last (as in the fourth line) from the definition (38) of μ .

Second, let $Y_k \subset \mathcal{Y}^k$ and consider $C_k(Y_k)$. We have:

$$\begin{aligned}
& \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(C_k(Y_k)|\tilde{y}^t) \mu(d\tilde{y}^t) \\
&= \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} \mu(d\tilde{y}^t), \\
&= \sum_{\tilde{y}_{(t-k+2):t}^t \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}_{t-k+2}^t} \Pi_{\tilde{y}_{t-k+2}^t \tilde{y}_{t-k+3}^t} \cdots \Pi_{\tilde{y}_{t-1}^t \tilde{y}_t^t} \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t}, \\
&= \sum_{y_{(t-k+2):t+1} \in Y_k} \sum_{\tilde{y}_{(t-k+2):t}^t \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}_{t-k+2}^t} \Pi_{\tilde{y}_{t-k+2}^t \tilde{y}_{t-k+3}^t} \cdots \Pi_{\tilde{y}_{t-1}^t \tilde{y}_t^t} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t}, \\
&= \sum_{y_{(t-k+2):t+1} \in Y_k} \pi_{y_{t-k+2}} \Pi_{y_{t-k+2} y_{t-k+3}} \cdots \Pi_{y_{t-1} y_t} \Pi_{y_t y_{t+1}} = \mu(C_k(Y_k)),
\end{aligned}$$

where the first equality comes from the definition (40) of μ_1 , the second from the fact that the integral is actually carried over a cylinder set of the form $C_{k-1}(\mathcal{Y}^{k-1})$ and from the definition (38) of μ , the third from the permutation of finite sums, the fourth from the fact that all terms in the sum over $\tilde{y}_{(t-k+2):t}^t \in \mathcal{Y}^{k-1}$ are zero except the one for $\tilde{y}_{(t-k+2):t}^t = y_{(t-k+2):t}$, and the last from the definition (38) of μ . This proves (45) on $\mathcal{C}_0$. Since cylinder sets are a generating family of $\mathcal{F}$ and a π -system, the equality (45) also holds on $\sigma(\mathcal{C}_0) = \mathcal{F}$.

We finally prove the following lemma. It justifies that in the main text we define μ_1 as $\mu_1(dy^{t+1}|\tilde{y}^t) = \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(dy^t)$.

Lemma 7. *For all $\tilde{y}^t \in \mathcal{Y}^\infty$ and for all $C \in \mathcal{C}_0$, we have:*

$$\mu_1(C|\tilde{y}^t) = \int_{y^{t+1} \in C} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})),$$

where $\delta_{\tilde{y}^t}$ is the Dirac mass in $\tilde{y}^t$.

Proof. Let $Y_k \subset \mathcal{Y}^k$ for some $k \geq 1$.

For $k = 1$, we have:

$$\begin{aligned}
\int_{y^{t+1} \in C_1(Y_1)} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) &= \int_{(y^t, y_{t+1}) \in \mathcal{Y}^\infty \times Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} \delta_{\tilde{y}^t}(dy^t), \\
&= \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} \int_{y^t \in \mathcal{Y}^\infty} \delta_{\tilde{y}^t}(dy^t), \\
&= \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} = \mu_1(C_1(Y_1)|\tilde{y}^t),
\end{aligned}$$

where the first equality comes from using $C_1(Y_1) = \mathcal{Y}^\infty \times Y_1$, the second from using Fubini's theorem (we consider σ -finite measure spaces and $(\tilde{y}^t, y_{t+1}) \in \mathcal{Y}^\infty \times Y_1 \mapsto \Pi_{\tilde{y}_t^t y_{t+1}} \delta_{\tilde{y}^t}(dy^t)$ is integrable), the third the property of Dirac mass, and the fourth the definition of μ_1 .

For $k \geq 2$, we have:

$$\int_{y^{t+1} \in C_k(Y_k)} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) \quad (46)$$

$$= \int_{(y^{t-k+1}, y_{(t-k+2):t+1}) \in \mathcal{Y}^\infty \times Y_k} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} \delta_{L^{(k-1)}(\tilde{y}^t)}(dy^{t-k+1}), \quad (47)$$

where $L^{(k-1)}(\cdot)$ is the k -iterate of the shift operator L : $L^{(k-1)}(\tilde{y}^t) = (\dots, \tilde{y}_{t-k}^t, \tilde{y}_{t-k+1}^t)$. To write equality (46), we used that $C_k(Y_k) = \mathcal{Y}^\infty \times Y_k$ and the fact that $\delta_{\tilde{y}^t}(Y_{k-1} \times C) = \sum_{y_{(t-k+2):t} \in Y_{k-1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t} \delta_{L^{(k-1)}(\tilde{y}^t)}(C)$ for all $C \in \mathcal{C}_0$ and $Y_{k-1} \subset \mathcal{Y}^{k-1}$. Hence the two measures coincide on $\mathcal{C}_0$ and are also σ -finite. Using Fubini and the property of a Dirac mass, we obtain from (46):

$$\begin{aligned} \int_{y^{t+1} \in C_k(Y_k)} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) &= \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}^{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}_{(t-k+2):t}^t}, \\ &= \mu_1(C_k(Y_k) | \tilde{y}^t), \end{aligned}$$

where the last equality comes from the definition of μ_1 . This concludes the proof. $\square$

A.5 Proof of Equality 5

We consider an agent whose individual preferences are represented by a utility function denoted by $V : \mathcal{Y}^\infty \rightarrow \mathbb{R}$. The allocation is subsumed. Let $y^t \in \mathcal{Y}^\infty$. The representation result of equation (4) becomes, after splitting the sum for $s = 0$ and $s \geq 1$:

$$V(y^t) = U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_s(dy^{t+s} | y^t). \quad (48)$$

Using the definition of μ_s given in Section 3.1 and Bayes' rule, we have, for all $s \geq 1$:

$$\mu_s(dy^{t+s} | y^t) = \int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_{s-1}(dy^{t+s} | y^{t+1}) \mu_1(dy^{t+1} | y^t). \quad (49)$$

In words, it means that the probability of transitioning from y^t to y^{t+s} in s periods is equal to the product of the probabilities of transitioning from y^t to y^{t+1} (in 1 period) and from y^{t+1} to y^{t+s} (in $s-1$ periods), summed over all possible histories y^{t+1} . Using (49), the expression of V in (48) becomes:

$$\begin{aligned} V(y^t) &= U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \\ &\quad \times \int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_{s-1}(dy^{t+s} | y^{t+1}) \mu_1(dy^{t+1} | y^t). \end{aligned}$$

Since all functions are integrable, we can use Fubini's theorem twice, to swap the order of the integrals and then the order of the sum and the integral on $y^{t+1} \in \mathcal{Y}^\infty$. We obtain:

$$V(y^t) = U(y^t) + \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} \left\{ \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_{s-1}(dy^{t+s} | y^{t+1}) \right\} \mu_1(dy^{t+1} | y^t). \quad (50)$$

The term between curly braces is $V(y^{t+1})$ using (4) applied to $t+1$. The previous equality becomes:

$$V(y^t) = U(y^t) + \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1} | y^t),$$

which gives the representation (5) using the notation $\mathbb{E}_{y^{t+1}} [V(y^{t+1}) | y^t] = \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1} | y^t)$ for the conditional expectation.

A.6 Proof of Proposition 3

Using the definition (7), the expression (8) of the H-SWF becomes:

$$\mathcal{W}^H = \int_{y^t \in \mathcal{Y}^\infty} p(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t),$$

which can be further simplified using the expression (6) of $\hat{V}$:

$$\begin{aligned} \mathcal{W}^H &= \int_{y^t \in \mathcal{Y}^\infty} p(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s} | \tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t), \\ I &= \int_{y^t \in \mathcal{Y}^\infty} p(y^t) \sum_{s=0}^{\infty} \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s} | \tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t). \end{aligned}$$

We assume absolute integrability of the discounted weighted payoff. Under this condition, Fubini's theorem permits us to permute the order of the sum and integrals:

$$\begin{aligned} \mathcal{W}^H &= \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \left[\int_{y^t \in \mathcal{Y}^\infty} p(y^t) \hat{\omega}(y^t, \hat{y}^{t+s}) \mu(dy^t) \right] U(\hat{y}^{t+s}) \\ &\quad \times \left[\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s} | \tilde{y}^t) \mu(d\tilde{y}^t) \right]. \end{aligned}$$

A straightforward extension of Lemma 6 for μ_s (for any $s \geq 1$) yields: $\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s} | \tilde{y}^t) \mu(d\tilde{y}^t) = \mu(d\hat{y}^{t+s})$. Using the definition (10) of the weights ω , we deduce:

$$\mathcal{W}^H = \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \omega(\hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu(d\hat{y}^{t+s}),$$

which proves Proposition 3.

As a final remark, observe that by splitting the sum over s into $s = 0$ and a sum for $s \geq$

1, we also obtain the following expression for $\mathcal{W}^H$: $\mathcal{W}^H = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t) \mu(dy^t) + \beta \mathcal{W}^H$.

B Competitive equilibrium

We provide a formal definition of a competitive equilibrium.

Definition 2 (Competitive equilibrium). A sequential competitive equilibrium is a collection of individual allocations $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$, of aggregate quantities $(K_t, L_t, Y_t)_{t \geq 0}$, of price processes $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t \geq 0}$, and of fiscal policies $(\tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t)_{t \geq 0}$, such that, for initial conditions and initial values of capital stock and public debt satisfying $K_{-1} + B_{-1} = \int_i a_{i,-1} \ell(di)$, we have:

1. given prices, the functions $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$ solve the agents' optimization program in equations (19)–(21);
2. financial, labor, and goods markets clear at all dates: for any $t \geq 0$, equation (24) holds;
3. the government's budget is balanced at all dates: equation (17) holds for all $t \geq 0$;
4. factor prices $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t \geq 0}$ are consistent with the condition (13) and post-tax definitions (16).

A steady-state competitive equilibrium is a competitive equilibrium in which the joint distribution of agents' decisions (c, l, a, ν) , aggregate quantities K, L, Y , prices $w, r, \tilde{w}, \tilde{r}$, and fiscal policy $(\tau^c, \tau^K, \tau, \kappa, B)$ are time-invariant.

C The Ramsey program 5

C.1 Reformulating the Ramsey program

We now reformulate the Ramsey problem. We define the following variables:

$$\tilde{a}_{i,t} := \frac{a_{i,t}}{1 + \tau_t^c}, \quad (51)$$

$$W_t := \frac{w_t}{1 + \tau_t^c}, \quad (52)$$

$$R_t := \frac{(1 + r_t)(1 + \tau_{t-1}^c)}{1 + \tau_t^c}, \quad (53)$$

which represent the asset choices in (51), the wage rate in (52), and the interest rate in (53). With this notation, the agent's budget and credit constraints become:

$$c_{i,t} + \tilde{a}_{i,t} = W_t(y_{i,t}l_{i,t})^{1-\tau_t} + R_t\tilde{a}_{i,t-1}, \quad (54)$$

$$\tilde{a}_{i,t} \geq -\frac{\bar{a}}{1 + \tau_t^c} := -\tilde{\bar{a}}. \quad (55)$$

Since taxes and prices are considered as given by agents, we can equivalently state their optimization program using the notation (51)–(53) and the constraints (54) and (55), rather than the original notation and the constraints (20) and (21). This modifies the Euler equations (22)–(23) as follows:

$$\begin{aligned} u'(c_{i,t}) &= \beta \mathbb{E}_t \left[R_{t+1} u'(c_{i,t+1}) \right] + \nu_{i,t}, \\ v'(l_{i,t}) &= (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}). \end{aligned}$$

We now turn to the government budget constraint. We further define:

$$\tilde{B}_t := \frac{B_t}{(1 + \tau_t^c)}, \quad (56)$$

$$\tilde{A}_t := \frac{A_t}{1 + \tau_t^c}, \quad (57)$$

and

$$\hat{B}_t := (1 + \tau_t^c) \tilde{B}_t - \tau_t^c \tilde{A}_t. \quad (58)$$

With these new definitions, the financial market clearing condition given by (32) holds, since we have $\tilde{A}_t = \int_i \tilde{a}(i) l(di)$.

Using the government budget constraint defined in (17), we have:

$$G_t + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t}l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t, s_t) + B_t.$$

Using the resource constraint $C_t + G_t + K_t = F(K_{t-1}, L_t, s_t) + K_{t-1}$, we obtain:

$$\begin{aligned} G_t + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t}l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} &= \\ \tau_t^c (F(K_{t-1}, L_t, s_t) - G_t - (K_t - K_{t-1})) + F(K_{t-1}, L_t, s_t) + B_t. \end{aligned}$$

Divide both sides of the equation above by $(1 + \tau_t^c)$ and obtain:

$$\begin{aligned} G_t + \frac{1 + r_t}{1 + \tau_t^c} B_{t-1} + \frac{w_t}{1 + \tau_t^c} \int_i (y_{i,t}l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1 + \tau_t^c} K_{t-1} &= \\ -\frac{\tau_t^c}{1 + \tau_t^c} (K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \frac{B_t}{1 + \tau_t^c}. \end{aligned}$$

Using the definitions (52), (53), and (56):

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1 + \tau_t^c} K_{t-1} = -\frac{\tau_t^c}{1 + \tau_t^c} (K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \tilde{B}_t.$$

We now substitute the expressions for K_t and K_{t-1} . From (56) and (57), we have

$$K_{t-1} = A_{t-1} - B_{t-1} = (1 + \tau_{t-1}^c)(\tilde{A}_{t-1} - \tilde{B}_{t-1}) \text{ and:}$$

$$\begin{aligned} G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t(1 + \tau_{t-1}^c)}{1 + \tau_t^c} (\tilde{A}_{t-1} - \tilde{B}_{t-1}) = \\ \frac{\tau_t^c(1 + \tau_{t-1}^c)}{1 + \tau_t^c} (\tilde{A}_{t-1} - \tilde{B}_{t-1}) + F(K_{t-1}, L_t, s_t) - \tau_t^c(\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t. \end{aligned}$$

Observe from (53) and (58) that $\frac{r_t(1 + \tau_{t-1}^c)}{1 + \tau_t^c} = R_t - \frac{1 + \tau_{t-1}^c}{1 + \tau_t^c}$ and $-\tau_t^c(\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t = \hat{B}_t$.

This yields:

$$\begin{aligned} G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - (1 + \tau_{t-1}^c))(\tilde{A}_{t-1} - \tilde{B}_{t-1}) = \\ F(K_{t-1}, L_t, s_t) + \hat{B}_t. \end{aligned}$$

Finally, using (58) in period $t - 1$ (i.e., $\hat{B}_{t-1} = (1 + \tau_{t-1}^c)\tilde{B}_{t-1} - \tau_{t-1}^c\tilde{A}_{t-1}$) we get:

$$\begin{aligned} G_t + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1)\tilde{A}_{t-1} + \hat{B}_{t-1} = \\ F(K_{t-1}, L_t, s_t) + \hat{B}_t. \end{aligned}$$

Since the public debt can be freely chosen by the planner, it is equivalent for the planner to choose $\hat{B}_t$ rather than B_t .

The reformulated Ramsey program. We reformulate the Ramsey program (59)–(65) using the variables $\tilde{a}_{i,t}$, W_t , R_t , $\tilde{A}_t$, $\hat{B}_t$ introduced in (51)–(53) and (57)–(58). The program can be expressed in post-tax prices R_t and W_t – taxes and pre-tax factor prices can be deduced from the allocation and the post-tax price definitions. The following proposition summarizes the reformulation of the Ramsey program.

Proposition 5. *The Ramsey program (59)–(65) can be rewritten as:*

$$\max_{(W_t, R_t, \tau_t, \hat{B}_t, \tilde{A}_t, K_t, L_t, (c_{i,t}, l_{i,t}, \tilde{a}_{i,t}, \nu_{i,t})_i)_{t \geq 0}} SWF_0, \quad (59)$$

$$G + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1) \tilde{A}_{t-1} + \hat{B}_{t-1} = F(K_{t-1}, L_t, s_t) + \hat{B}_t, \quad (60)$$

$$\text{for all } i \in \mathcal{I}: c_{i,t} + \tilde{a}_{i,t} = W_t (y_{i,t} l_{i,t})^{1-\tau_t} + R_t \tilde{a}_{i,t-1}, \quad (61)$$

$$\tilde{a}_{i,t} \geq -\tilde{a}, \quad \nu_{i,t}(\tilde{a}_{i,t} + \tilde{a}) = 0, \quad \nu_{i,t} \geq 0, \quad (62)$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[R_{t+1} u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (63)$$

$$v'(l_{i,t}) = (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (64)$$

$$K_t + \hat{B}_t = \tilde{A}_t = \int_i \tilde{a}_{i,t} \ell(di), \quad L_t = \int_i y_{i,t} l_{i,t} \ell(di). \quad (65)$$

C.2 The Lagrangian, the FOCs of the Ramsey program and equilibrium definitions

The Lagrangian associated with the Ramsey program (59)–(65) can be written as:

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u'(c_{i,t}) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \lambda_{l,i,t} \left(v'(l_{i,t}) - (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}) \right) \ell(di) \\ & - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left(G_t + (1 - \delta) \hat{B}_{t-1} + (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1} \ell(di) + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) - Y_t - \hat{B}_t \right), \end{aligned}$$

where the value of $\nu_{i,t}$ is given by the complementary slackness conditions (62) and (63), and where we have:

$$c_{i,t} = -\tilde{a}_{i,t} + R_t \tilde{a}_{i,t-1} + W_t (y_{i,t} l_{i,t})^{1-\tau_t}, \quad (66)$$

$$Y_t = \left(\int_i \tilde{a}_{i,t-1} \ell(di) - \hat{B}_{t-1} \right)^\alpha \left(\int_i y_{i,t} l_{i,t} \ell(di) \right)^{1-\alpha}. \quad (67)$$

Consequently, the instruments are: $\tilde{a}_{i,t}$, $l_{i,t}$, W_t , R_t , τ_t , and $\hat{B}_t$. Using the two previous equations to substitute $c_{i,t}$ and Y_t , the planner's program is:

$$\max_{(W_t, R_t, \tau_t, \hat{B}_t, (l_{i,t}, \tilde{a}_{i,t})_i)_{t \geq 0}} \mathcal{L}.$$

We now provide the first-order conditions of the planner, and we present an alternative interpretation of the Lagrangian in the next section.

FOC with respect to public debt $\hat{B}_t$.

$$\mu_t = \beta \mathbb{E}_t [(1 + \tilde{r}_{t+1}) \mu_{t+1}]. \quad (68)$$

FOC with respect to saving choices $\tilde{a}_{i,t}$. We define the marginal social value of liquidity for agent i at date t as:

$$\psi_{i,t} := \omega_{i,t} u'(c_{i,t}) - (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1} - \lambda_{l,i,t} (1 - \tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t}) u''(c_{i,t}), \quad (69)$$

and $\hat{\psi}_{i,t} := \psi_{i,t} - \mu_t$ as the marginal social value of liquidity net of the planner's resource cost. Using (68), we obtain:

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t [R_{t+1} \hat{\psi}_{i,t+1}]. \quad (70)$$

FOC with respect to labor supply $l_{i,t}$. We define:

$$\psi_{l,i,t} := \omega_{i,t} v'(l_{i,t}) + \lambda_{l,i,t} v''(l_{i,t}).$$

The FOC with respect to labor supply $l_{i,t}$ is:

$$\psi_{l,i,t} = (1 - \tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \hat{\psi}_{i,t} + \mu_t F_{L,t} y_{i,t} - (1 - \tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \lambda_{l,i,t} \tau_t \frac{u'(c_{i,t})}{l_{i,t}}.$$

FOC with respect to the wage rate W_t .

$$0 = \int_j (y_{j,t} l_{j,t})^{1-\tau_t} (\hat{\psi}_{j,t} + \lambda_{l,j,t} (1 - \tau_t) u'(c_{j,t}) / l_{j,t}) \ell(dj).$$

FOC with respect to the interest rate R_t .

$$0 = \int_j (\hat{\psi}_{j,t} \tilde{a}_{t-1}^j + \lambda_{c,j,t-1} u'(c_{j,t})) \ell(dj). \quad (71)$$

FOC with respect to progressivity τ_t .

$$\begin{aligned} 0 = & \int_j (y_{j,t} l_{j,t})^{1-\tau_t} (\hat{\psi}_{j,t} + \lambda_{l,j,t} (1 - \tau_t) (u'(c_{j,t}) / l_{j,t})) \ln(y_{j,t} l_{j,t}) \ell(dj) \\ & + \int_j \lambda_{l,j,t} (y_{j,t} l_{j,t})^{1-\tau_t} (u'(c_{j,t}) / l_{j,t}) \ell(dj). \end{aligned} \quad (72)$$

A *Ramsey allocation* is a competitive equilibrium in which the fiscal policy and allocations satisfy the first-order conditions of the planner (68)-(72).

A *Ramsey steady state* is a Ramsey allocation in which fiscal policy $(\tau^c, \tau^K, \tau, \kappa, B)$, aggregate variables (K, A, B) and prices r, w are constant.

C.3 Expression of the Lagrangian using a public finance representation

The Lagrangian can be written as:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (\mathcal{W}_t + \mu_t \mathcal{B}_t),$$

with:

$$\begin{aligned} \mathcal{W}_t &:= \int_i (\omega_{i,t}(u(c_{i,t}) - v(l_{i,t})) - (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u'(c_{i,t}) \\ &\quad - \lambda_{l,i,t} (v'(l_{i,t}) - (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}))) \ell(di), \\ \mathcal{B}_t &:= Y_t - \hat{B}_t - G_t - (1 - \delta) \hat{B}_{t-1} - (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1} \ell(di) - W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di). \end{aligned}$$

The term $\mathcal{B}_t$ collects the budget constraint of the government. The term $\mathcal{W}_t$ collects the welfare contribution of individual choices, including the general-equilibrium effects captured by the Lagrange multipliers on household optimal choices, $\lambda_{c,i,t}$ and $\lambda_{l,i,t}$. If these multipliers were zero, then $\mathcal{W}_t$ would reduce to the weighted sum of individual utilities.

Considering an instrument I_k in period k (I_k can be public debt, the interest rate, the labor tax, or labor-tax progressivity), the FOC of the Lagrangian with respect to I_k implies:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \frac{d\mathcal{B}_t}{dI_k} + \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int \frac{\partial \mathcal{W}_t}{\partial I_k} \ell(di) = -\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int \frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \frac{\partial c_{i,t}}{\partial I_k} \ell(di). \quad (73)$$

Since W_t, R_t, τ_t only affect the current value of welfare and the budget constraint (for $I_t \in \{W_t, R_t, \tau_t\}$, $\frac{\partial \mathcal{W}_t}{\partial I_k} = \frac{\partial \mathcal{B}_t}{\partial I_k} = 0$ if $k \neq t$), we have for any $I_t \in \{W_t, R_t, \tau_t\}$:

$$\mu_t = \int \frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \frac{-\frac{\partial c_{i,t}}{\partial I_t}}{\frac{d\mathcal{B}_t}{dI_t} + \frac{1}{\mu_t} \frac{\partial \mathcal{W}_t}{\partial I_t}} \ell(di). \quad (74)$$

It is then straightforward to compute the partial derivatives and verify that they yield the same expressions as in Section C.2. For instance, we have $\frac{\partial \mathcal{W}_t}{\partial c_{i,t}} = \psi_{i,t}$ and when the fiscal instrument is the post-tax rate R_t : $\frac{d\mathcal{B}_t}{dR_t} = -\int_i \tilde{a}_{i,t-1} \ell(di)$, $\frac{\partial \mathcal{W}_t}{\partial R_t} = \int_i R_t \lambda_{c,i,t-1} u'(c_{i,t})$, and $\frac{\partial c_{i,t}}{\partial R_t} = \tilde{a}_{i,t-1}$. Then, (74) becomes:

$$\mu_t = \int \psi_{i,t} \frac{-\tilde{a}_{i,t-1}}{-\int_i \tilde{a}_{i,t-1} \ell(di) + \frac{1}{\mu_t} \int_i R_t \lambda_{c,i,t-1} u'(c_{i,t})} \ell(di),$$

which can be written as $\int (\psi_{i,t} - \mu_t) \tilde{a}_{i,t-1} \ell(di) + \int_i R_t \lambda_{c,i,t-1} u'(c_{i,t}) = 0$, which is the FOC (71) with $\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t$. The same derivations can be obtained for W_t and τ_t .

The instrument $\hat{B}_t$ is a fiscal instrument that affects current and future budget constraints but not welfare: $\frac{\partial \mathcal{W}_t}{\partial \hat{B}_k} = 0$, for all k and $\frac{\partial \mathcal{B}_t}{\partial \hat{B}_t} = -1$, $\frac{\partial \mathcal{B}_{t+1}}{\partial \hat{B}_t} = 1 + \tilde{r}_{t+1}$, and $\frac{\partial \mathcal{B}_{\tau}}{\partial \hat{B}_t} = 0$ if

$\tau \notin \{t, t+1\}$. As a consequence, the FOC (73) simplifies into $\mathbb{E}_0[\beta^t \mu_t \frac{dB_t}{dB_t} + \beta^{t+1} \mu_{t+1} \frac{dB_{t+1}}{dB_t}] = 0$, or $-\mu_t + \beta \mathbb{E}_t[\mu_{t+1}(1 + \tilde{r}_{t+1})] = 0$, which is the FOC (68).

D Truncating the model and identifying G-SWF weights

D.1 The truncated model

The key step of the aggregation consists of assigning the same wealth and allocation to all agents sharing the same idiosyncratic history over the recent past. The recent past is characterized by a number of periods, called the *truncation length* and denoted N , which is a parameter of the model. This N -period history will be referred to as a *truncated history*. For a history $y^t = \{\dots, y_{t-N}^t, y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}$, this corresponds to the N -length vector denoted $y^N := \{y_{t-N}^t, y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}$. To sum up, we can represent the truncated history of an agent i whose idiosyncratic history is y^t as:

$$y^t = \underbrace{\{\dots, y_{t-N-2}^t, y_{t-N-1}^t, y_{t-N}^t\}}_{\xi_{y^N}} \underbrace{\{y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}}_{=y^N},$$

where the parameter ξ_{y^N} captures the residual heterogeneity for the truncated history y^N , and y_{t-k}^t represents the idiosyncratic variable (at date t) k periods in the past. The method to compute the set of parameters $(\xi_{y^N})_{y^N}$ will be discussed further below. In what follows, we will discuss the various elements needed to apply the aggregation procedure.

First, we need to compute the measure of agents with the same history y^N . An agent with history $\tilde{y}^N$ at $t-1$ will have a different truncated history in period t depending on the realization of the idiosyncratic variable at date t . The probability of transitioning from truncated history $\tilde{y}^N$ to truncated history y^N will be denoted by $\Pi_{\tilde{y}^N y^N}$ (with $\sum_{y^N \in \mathcal{Y}^N} \Pi_{\tilde{y}^N y^N} = 1$) and can be computed from the transition probabilities for the productivity process as:

$$\Pi_{\tilde{y}^N y^N} = 1_{y^N \succeq \tilde{y}^N} \Pi_{\tilde{y}_0^N y_0^N} \geq 0,$$

where $1_{y^N \succeq \tilde{y}^N}$ is equal to 1 if y^N is a possible continuation of $\tilde{y}^N$, and 0 otherwise. With these elements, we can compute the share of agents with truncated history y^N as S_{t, y^N} . This element is:

$$S_{t, y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} S_{t-1, \tilde{y}^N} \Pi_{\tilde{y}^N y^N}, \quad (75)$$

where the initial shares $(S_{-1, y^N})_{y^N \in \mathcal{Y}^N}$ are given with $\sum_{y^N \in \mathcal{Y}^N} S_{-1, y^N} = 1$.

The model aggregation assigns to each truncated history the average choices (for consumption, saving, or labor supply) of agents who share that history. For any individual variable $X_t(y^t, s^t)$, let X_{t, y^N} denote its average among agents whose last N idiosyncratic

states are y^N :

$$X_{t,y^N} = \frac{1}{S_{t,y^N}} \sum_{y^t \in \mathcal{Y}^{t+1} | (y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t) = y^N} X_t(y^t, s^t) \mu_t(y^t), \quad (76)$$

where we recall that $\mu_t(y^t)$ is the measure of agents with history y^t . We apply this definition to consumption, savings, labor supply, and the borrowing-constraint multiplier, obtaining c_{t,y^N} , $\tilde{a}_{t,y^N}$, l_{t,y^N} , and ν_{t,y^N} , respectively. Beginning-of-period wealth requires one additional step because agents with the same truncated history at date t may have arrived from different truncated histories at date $t-1$. Specifically, this variable consists of the wealth of all agents with history y^N in period t but with any other possible history at $t-1$. Formally, the beginning-of-period wealth $\tilde{a}_{t,y^N}$ for truncated history y^N is:

$$\tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (77)$$

We now define the various “ ξ s”. First, we define $\xi_{y^N}^{u,0}$ as:

$$\sum_{y^t \in \mathcal{Y}^{t+1} | (y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t) = y^N} u(c_t(y^t)) = \xi_{y^N}^{u,0} u\left(\sum_{y^t \in \mathcal{Y}^{t+1} | (y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t) = y^N} c_t(y^t)\right),$$

or compactly as:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u(c_{i,t}) = \xi_{y^N}^{u,0} u(c_{t,y^N}). \quad (78)$$

The quantity $\xi_{y^N}^{u,0}$ reflects that aggregating utility levels is not equal to the utility of aggregated consumption. This comes from a combination of two factors. First, there is heterogeneity of consumption among the population of agents with the truncated history y^N , due to their history prior to date $t-N$. Second, the utility function is not affine in general.

The same procedure applied to the other variables for the Ramsey problem (59)–(65) yields:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} v(l_{i,t}) := \xi_{y^N}^{v,0} v(l_{t,y^N}), \quad (79)$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u'(c_{i,t}) := \xi_{y^N}^{u,1} u'(c_{t,y^N}), \quad (80)$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} := \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t}. \quad (81)$$

We can now proceed with the aggregation of the full-fledged model. First, the aggregation of individual budget constraints (54) yields:

$$c_{t,y^N} + \tilde{a}_{t,y^N} = W_t \xi_{y^N}^y (l_{t,y^N} y_0^N)^{1-\tau_t} + R_t \tilde{a}_{t,y^N}, \text{ for } y^N \in \mathcal{Y}^{\infty N}. \quad (82)$$

The aggregation of Euler equations for consumption (63) and labor (64) yields:

$$\xi_{y^N}^{u,E} u'(c_{t,y^N}) = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \Pi_{t+1,y^N \tilde{y}^N} \xi_{\tilde{y}^N}^{u,E} u'(c_{t+1,\tilde{y}^N}) \right] + \nu_{t,y^N}, \quad (83)$$

$$\xi_{y^N}^{v,1} v'(l_{t,y^N}) := (1 - \tau_t) W_t \xi_{y^N}^y (l_{t,y^N} y_0^N)^{1-\tau_t} \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}), \quad (84)$$

where the coefficients $(\xi_{y^N}^{u,E})_{y^N}$ for the consumption Euler equations ensure that the aggregate Euler equations yield Euler equations with aggregate consumption levels. In other words, the $(\xi_{y^N}^{u,E})_{y^N}$ are determined such that the aggregated consumption levels (for truncated histories) satisfy the consumption Euler equation (83). These coefficients are necessary because Euler equations involve nonlinear marginal utilities. The same idea applies to the coefficients $(\xi_{y^N}^{v,1})_{y^N}$ for the FOC on labor.

Finally, the market clearing conditions can be expressed as:

$$K_t + \hat{B}_t = \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \tilde{a}_{t,y^N}, \quad L_t = \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} y_{y^N} l_{t,y^N}. \quad (85)$$

Equations (82)–(85) exactly characterize the dynamics of the aggregated variables c_{t,y^N} , $\tilde{a}_{t,y^N}$, l_{t,y^N} , and ν_{t,y^N} , as well as aggregate quantities K_t , $\hat{B}_t$, and L_t .

Steady state and computation of the ξ s. Steady-state allocations allow us to compute the parameters ξ s as follows. We compute the policy functions and wealth distribution of the Bewley model, and identify the set of credit-constrained histories, denoted $\mathcal{C}$. Aggregation equations (76) and (77) can then be used to aggregate (steady-state) allocations c_{y^N} , $\tilde{a}_{y^N}$, l_{y^N} , and ν_{y^N} . We then invert the consumption Euler equations (83) to deduce the preference parameters $(\xi_{y^N}^{u,E})_{y^N}$. The other ξ s are computed explicitly by equations (78), (79), (80), (81), and (84).

The truncated model in the presence of aggregate shocks. We state two further assumptions that enable us to use the truncation method in the presence of aggregate shocks, yielding the so-called truncated model.

Assumption B. *We make the following two assumptions:*

1. The preference parameters $(\xi_{y^N})_{y^N}$ remain constant and equal to their steady-state values.
2. The set of credit-constrained histories, denoted by $\mathcal{C} \subset \mathcal{Y}^{\infty N}$, is time-invariant.

Finally, two properties are worth mentioning. First, a straightforward consequence of the construction of the ξ s is that the steady-state allocations of the initial and truncated models are identical. Second, as the truncation length N becomes longer, truncated allocations (in the presence of aggregate shocks) can be shown to converge to those of

the full-fledged equilibrium. Section 5 shows that from a quantitative standpoint, the ξ s efficiently capture the heterogeneity within truncated histories, even when the truncation length remains limited.

D.2 Ramsey program

Program formulation. The finite state-space representation of the truncated model allows us to solve for the Ramsey program in the presence of aggregate shocks.¹⁸ Let $(\omega_y)_{y \in \mathcal{Y}}$ denote the period weights associated with each productivity level. The Ramsey program in the truncated economy can be written as follows:

$$\max_{(W_t, R_t, \tilde{w}_t, \tilde{r}_t, \tau_t^c, \tau_t^K, \tau_t, \kappa_t, \hat{B}_t, G_t, K_t, L_t, (c_{t,y^N}, l_{t,y^N}, \tilde{a}_{t,y^N}, \nu_{t,y^N})_{y^N})_{t \geq 0}} W_0, \quad (86)$$

where $W_0 := \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \omega_{y^N} (\xi_{y^N}^{u,0} u(c_{t,y^N}) - \xi_{y^N}^{v,0} v(l_{t,y^N}) + u_G(G_t)) \right]$ and subject to aggregate Euler equations (83) and (84), aggregate budget constraint (82), aggregate market clearing conditions (85), credit constraints $\tilde{a}_{t,y^N} \geq -\tilde{a}$, as well as the government budget constraint (60), which is already present in the full-fledged Ramsey program.

First-order conditions. We define the net social value of liquidity of history y^N as in (69):

$$\begin{aligned} \hat{\psi}_{t,y^N} &= \omega_{y^N} \xi_{y^N}^{u,0} u'(c_{t,y^N}) - \mu_t \\ &\quad - \left(\lambda_{c,t,y^N} \xi_{y^N}^{u,E} - R_t \tilde{\lambda}_{c,t,y^N} \xi_{y^N}^{u,E} - \lambda_{l,t,y^N} \xi_{y^N}^{y} (1 - \tau_t) W_t (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t} \xi_{y^N}^{u,1} \right) u''(c_{t,y^N}). \end{aligned} \quad (87)$$

FOC with respect to $\tilde{a}_{t,y^N}$:

$$\hat{\psi}_{t,y^N} = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \Pi_{t,y^N \tilde{y}^N} \hat{\psi}_{t+1,\tilde{y}^N} \right] \text{ if } \nu_{y^N} = 0 \text{ and } \lambda_{c,t,y^N} = 0 \text{ otherwise.} \quad (88)$$

FOC with respect to l_{t,y^N} :

$$\begin{aligned} \frac{\omega_{y^N} \xi_{y^N}^{v,0} v'(l_{t,y^N}) + \lambda_{l,t,y^N} \xi_{y^N}^{v,1} v''(l_{t,y^N})}{(1 - \tau_t) W_t \xi_{y^N}^y (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t}} &= \hat{\psi}_{t,y^N} - \lambda_{l,t,y^N} \tau_t \xi_{y^N}^{u,1} (u'(c_{t,y^N}) / l_{t,y^N}) \\ &\quad + \mu_t (1 - \alpha) \frac{Y_t}{(1 - \tau_t) W_t \xi_{y^N}^y (y_0^N)^{-\tau_t} l_{t,y^N}^{-\tau_t} L_t}. \end{aligned} \quad (89)$$

¹⁸Our method involves deriving the FOCs of the truncated model, rather than truncating the FOCs of the full-fledged Ramsey model. This ensures numerical stability, as the truncated model is "well-defined" for the fiscal policy under consideration by construction.

FOC with respect to W_t :

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1-\tau_t) \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}) \right) = 0. \quad (90)$$

FOC with respect to R_t :

$$\sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \left(\hat{\psi}_{t,y^N} \tilde{a}_{t,y^N} + \tilde{\lambda}_{c,t,y^N} \xi_{y^N}^{u,E} u'(c_{t,y^N}) \right) = 0. \quad (91)$$

FOC with respect to τ_t :

$$\begin{aligned} & \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1-\tau_t) \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}) \right) \ln \left(l_{t,y^N} y_{y^N} \right) \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \\ &= - \sum_{y^N \in \mathcal{Y}^{\infty N}} S_{t,y^N} \lambda_{l,t,y^N} \xi_{y^N}^y (l_{t,y^N} y_{y^N})^{1-\tau_t} \xi_{y^N}^{u,1} (u'(c_{t,y^N})/l_{t,y^N}). \end{aligned} \quad (92)$$

FOC with respect to $\hat{B}_t$:

$$\mu_t = \beta \mathbb{E} \left[\mu_{t+1} \left(1 + \alpha \frac{Y_{t+1}}{K_t} - \delta \right) \right]. \quad (93)$$

We must furthermore have:

$$\tilde{\lambda}_{c,t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \lambda_{c,t-1,\tilde{y}^N}, \quad (94)$$

$$\tilde{a}_{t,y^N} \geq 0 \text{ and } \tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (95)$$

D.3 Matrix expression

In this section, we provide closed-form formulas for preference multipliers ξ s (Section D.1) and the weights ω s. We start with some notation:

$\circ$ is the Hadamard product, $\otimes$ is the Kronecker product, $\times$ is the usual matrix product.

For any vector V , we denote by $\text{diag}(V)$ the diagonal matrix with V on the diagonal.

The matrix representation stacks the steady-state equations in a form that is convenient to solve the system. We index truncated histories by y^N : the precise ordering is irrelevant as long as it is used consistently. The main benefit of this matrix notation is to allow us to compute the coefficient vectors (ξ_{y^N}) and (ω_{y^N}) from a linear system.

D.3.1 A closed-form formula for the ξ s

Let $\mathbf{S} = (S_{y^N})_{y^N}$ be the N_{tot} -vector of steady-state history sizes (where N_{tot} is the number of truncated histories). Similarly, let $\mathbf{a}$, $\mathbf{c}$, $\mathbf{l}$, $\boldsymbol{\nu}$, $u'(\mathbf{c})$, $\mathbf{v}'(\mathbf{l})$, $\mathbf{u}''(\mathbf{c})$, $\mathbf{v}''(\mathbf{l})$ be the N_{tot} -

vectors of end-of-period wealth, consumption, labor supply, Lagrange multipliers, marginal utilities, and derivatives of the marginal utility, respectively. These vectors are known from the steady-state equilibrium of the Bewley model. Each element is defined as the truncation of the relevant variable computed using equation (76). We also define by $\mathbf{y} = (y_0^N)_{y^N}$ the N_{tot} -vector of current productivity levels of truncated histories, and by $\mathbf{P}$ the diagonal matrix having 1 on the diagonal at y^N if and only if the history y^N is not credit-constrained (i.e., $\nu_{y^N} = 0$), and 0 otherwise. Finally, $\mathbf{I}$ is the $(N_{tot} \times N_{tot})$ -identity matrix, and $\mathbf{\Pi}$ is the transition matrix across truncated histories.

Writing (75), (82) and credit constraints at the steady state yield, respectively:

$$\mathbf{S} = \mathbf{\Pi} \mathbf{S}, \quad (96)$$

$$\mathbf{S} \circ \mathbf{c} + \mathbf{S} \circ \tilde{\mathbf{a}} = R \mathbf{\Pi} (\mathbf{S} \circ \tilde{\mathbf{a}}) + W \mathbf{S} \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau}, \quad (97)$$

$$(\mathbf{I} - \mathbf{P}) \tilde{\mathbf{a}} = \mathbf{0}_{N_{tot} \times 1}. \quad (98)$$

The Euler equation for consumption in (83) becomes:

$$\boldsymbol{\xi}^{u,E} \circ u'(\mathbf{c}) = \beta R \mathbf{\Pi}^\top (\boldsymbol{\xi}^{u,E} \circ u'(\mathbf{c})) + \boldsymbol{\nu},$$

where the transpose matrix $\mathbf{\Pi}^\top$ implies expectations about next-period histories. Equivalently:

$$\mathbf{D}_{u'(\mathbf{c})} \boldsymbol{\xi}^{u,E} = \beta R \mathbf{\Pi}^\top \mathbf{D}_{u'(\mathbf{c})} \boldsymbol{\xi}^{u,E} + \boldsymbol{\nu},$$

where $\mathbf{D}_{\mathbf{x}}$ stands for the diagonal matrix with the vector $\mathbf{x}$ on the diagonal. Finally:

$$\boldsymbol{\xi}^{u,E} = \left[(\mathbf{I} - \beta R \mathbf{\Pi}^\top) \mathbf{D}_{u'(\mathbf{c})} \right]^{-1} \boldsymbol{\nu}. \quad (99)$$

From the FOC on labor in (84), we obtain:

$$\boldsymbol{\xi}^{v,1} = (1 - \tau) W (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \boldsymbol{\xi}^y \circ \boldsymbol{\xi}^{u,1} \circ u'(\mathbf{c}) ./ (\mathbf{l} \circ v'(\mathbf{l})). \quad (100)$$

The equations (78)–(81) yield:

$$\boldsymbol{\xi}^{u,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u(c_{i,t})}{u(c_{t,y^N})}, \quad \boldsymbol{\xi}^{u,1} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u'(c_{i,t})}{u'(c_{t,y^N})}, \quad (101)$$

$$\boldsymbol{\xi}^{v,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} v(l_{i,t})}{v(l_{t,y^N})}, \quad \boldsymbol{\xi}^y = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} (y_{i,t} l_{i,t})^{1-\tau}}{(y_0^N l_{t,y^N})^{1-\tau}}. \quad (102)$$

D.3.2 Matrix expressions for the FOCs

We define the following variables: $\bar{\boldsymbol{\lambda}}_l := \mathbf{S} \circ \boldsymbol{\lambda}_l$, $\bar{\boldsymbol{\psi}} := \mathbf{S} \circ \hat{\boldsymbol{\psi}}$, $\bar{\mathbf{\Pi}} := \mathbf{S} \circ \mathbf{\Pi}^\top \circ (1/\mathbf{S})$, $\bar{\boldsymbol{\omega}} := \mathbf{S} \circ \boldsymbol{\omega}$, $\bar{\boldsymbol{\lambda}}_c := \mathbf{S} \circ \boldsymbol{\lambda}_c$, $\tilde{\boldsymbol{\xi}}^{u,1} := \boldsymbol{\xi}^{u,1} ./ \mathbf{l}$, $\tilde{\boldsymbol{\xi}}^{v,1} := \boldsymbol{\xi}^{v,1} ./ ((1 - \tau) W \boldsymbol{\xi}^y \circ \mathbf{y}^{1-\tau} \circ \mathbf{l}^{-\tau})$, and $\tilde{\boldsymbol{\xi}}^{v,0} := \boldsymbol{\xi}^{v,0} ./ ((1 - \tau) W \boldsymbol{\xi}^y \circ \mathbf{y}^{1-\tau} \circ \mathbf{l}^{-\tau})$. Notice that $\mathbf{S} \circ \bar{\boldsymbol{\lambda}}_c = \mathbf{\Pi} \bar{\boldsymbol{\lambda}}_c$. The FOCs (87)–(93)

become:

$$\bar{\psi} = \bar{\omega} \circ \xi^{u,0} \circ u'(c) - \mu S \quad (103)$$

$$- \left(\bar{\lambda}_c \circ \xi^{u,E} - R\Pi\bar{\lambda}_c \circ \xi^{u,E} - (1-\tau)W\bar{\lambda}_l \circ \xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \right) \circ u''(c),$$

$$P\bar{\psi} = \beta RP\bar{\Pi}\bar{\psi}, \quad (104)$$

$$(I - P)\bar{\lambda}_c = 0, \quad (105)$$

$$\left(\xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = -(1-\tau) \left(\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l, \quad (106)$$

$$\tilde{a}^\top \bar{\psi} = - \left(\xi^{u,E} \circ u'(c) \right)^\top \Pi\bar{\lambda}_c, \quad (107)$$

$$\bar{\omega} \circ \tilde{\xi}^{v,0} \circ v'(l) + \bar{\lambda}_l \circ \tilde{\xi}^{v,1} \circ v''(l) = \bar{\psi} - \tau \tilde{\xi}^{u,1} \circ u'(c) \circ \bar{\lambda}_l + \mu F_L S. / ((1-\tau)W\xi^y \circ y^{-\tau} \circ l^{-\tau}), \quad (108)$$

$$\left(\ln(y \circ l) \circ \xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = - \left((1 + (1-\tau) \ln(y \circ l)) \circ \xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l. \quad (109)$$

D.3.3 Solving the system

Equation (108) yields:

$$\begin{aligned} D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)} \bar{\lambda}_l &= \mu F_L S. / ((1-\tau)W\xi^y \circ y^{-\tau} \circ l^{-\tau}) + \bar{\psi} - D_{\tilde{\xi}^{v,0} \circ v'(l)} \bar{\omega}, \\ \bar{\lambda}_l &= M_0 \bar{\omega} + M_1 \bar{\psi} + \mu V_0, \end{aligned} \quad (110)$$

with: $M_0 := -M_1 D_{\tilde{\xi}^{v,0} \circ v'(l)}$, $M_1 := D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)}^{-1}$, and $V_0 := F_L M_1 S. / ((1-\tau)W\xi^y \circ y^{-\tau} \circ l^{-\tau})$.

Equation (103) then implies:

$$\bar{\psi} = \hat{M}_0 \bar{\omega} + \hat{M}_1 \bar{\lambda}_c + \hat{M}_2 \bar{\lambda}_l - \mu S, \quad (111)$$

with: $\hat{M}_0 := D_{\xi^{u,0} \circ u'(c)}$, $\hat{M}_1 := -D_{\xi^{u,E} \circ u''(c)}(I - R\Pi)$, $\hat{M}_2 := (1-\tau)W D_{\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u''(c)}$.

We obtain using (111) and (110):

$$\bar{\psi} = M_3 \bar{\omega} + M_4 \bar{\lambda}_c + \mu V_1, \quad (112)$$

where $M_2 := I - \hat{M}_2 M_1$, $M_3 := M_2^{-1}(\hat{M}_0 + \hat{M}_2 M_0)$, $M_4 := M_2^{-1} \hat{M}_1$, $V_1 := M_2^{-1}(\hat{M}_2 V_0 - S)$.

Furthermore, equations (104), (105), and (112) imply:

$$\bar{\lambda}_c = M_5 \bar{\omega} + \mu V_2, \quad (113)$$

where $\tilde{R}_5 := -((I - P) + P(I - \beta R\Pi)M_4)^{-1}P(I - \beta R\Pi)$, $M_5 := \tilde{R}_5 M_3$, and $V_2 := \tilde{R}_5 V_1$.

Substituting (112) and (113) into (107), we deduce:

$$\mu = -\mathbf{L}_0 \bar{\omega}, \quad (114)$$

where $\mathbf{C}_1 := \tilde{\mathbf{a}}^\top (\mathbf{V}_1 + \mathbf{M}_4 \mathbf{V}_2) + (\boldsymbol{\xi}^{u,E} \circ u'(\mathbf{c}))^\top \mathbf{\Pi} \mathbf{V}_2$ and $\mathbf{L}_0 := (\tilde{\mathbf{a}}^\top (\mathbf{M}_3 + \mathbf{M}_4 \mathbf{M}_5) + (\boldsymbol{\xi}^{u,E} \circ u'(\mathbf{c}))^\top \mathbf{\Pi} \mathbf{M}_5) / \mathbf{C}_1$.

We deduce from (112) and (113):

$$\bar{\boldsymbol{\lambda}}_c = (\mathbf{M}_5 - \mathbf{V}_2 \mathbf{L}_0) \bar{\omega}, \quad (115)$$

$$\bar{\boldsymbol{\psi}} = \mathbf{M}_6 \bar{\omega}, \quad (116)$$

and from (110):

$$\bar{\boldsymbol{\lambda}}_l = \hat{\mathbf{M}}_6 \bar{\omega}. \quad (117)$$

We have defined $\hat{\mathbf{M}}_6 := \mathbf{M}_0 + \mathbf{M}_1 \mathbf{M}_6 - \mathbf{V}_0 \mathbf{L}_0$ and $\mathbf{M}_6 := \mathbf{M}_3 + \mathbf{M}_4 (\mathbf{M}_5 - \mathbf{V}_2 \mathbf{L}_0) - \mathbf{V}_1 \mathbf{L}_0$.

Constructing the constraints. The constraint in equation (109) after substituting the expressions (116) of $\bar{\boldsymbol{\psi}}$ and (117) of $\bar{\boldsymbol{\lambda}}_l$, becomes:

$$\tilde{\mathbf{L}}_1 \bar{\omega} = 0, \quad (118)$$

where:

$$\begin{aligned} \tilde{\mathbf{L}}_1 &:= \left(\ln(\mathbf{y} \circ \mathbf{l}) \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \right)^\top \mathbf{M}_6 \\ &\quad + \left((1 + (1 - \tau) \ln(\mathbf{y} \circ \mathbf{l})) \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\mathbf{c}) \right)^\top \hat{\mathbf{M}}_6. \end{aligned}$$

The constraint (106) after substituting the expressions (116) of $\bar{\boldsymbol{\psi}}$ and (117) of $\bar{\boldsymbol{\lambda}}_l$, becomes:

$$\tilde{\mathbf{L}}_2 \bar{\omega} = 0, \quad (119)$$

where:

$$\tilde{\mathbf{L}}_2 := \left(\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \right)^\top \mathbf{M}_6 + (1 - \tau) \left(\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\mathbf{c}) \right)^\top \hat{\mathbf{M}}_6.$$

The two constraints imposed on the history weights are $\tilde{\mathbf{L}}_1 \bar{\omega} = 0$ and $\tilde{\mathbf{L}}_2 \bar{\omega} = 0$. These restrictions are written for a vector of length N_{tot} , whereas Definition 1 uses a productivity-state weight vector of length Y . We therefore introduce matrices that map history-level weights into productivity-level weights. The $N_{tot} \times Y$ -matrix $\mathbf{R}_0$ maps a

Y -vector into an N_{tot} -vector (where $\mathbf{1}_{S_y} \in \mathbb{R}^{S_y}$ is an S_y -vector of 1):

$$\mathbf{R}_0 := \begin{bmatrix} \mathbf{1}_{S_{y_1}} & 0 & 0 & 0 \\ 0 & \mathbf{1}_{S_{y_2}} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & S_{y_Y} \end{bmatrix},$$

and the $N_{tot} \times Y$ -matrix $\mathbf{R}_1 := \mathbf{D}_S \mathbf{R}_0$ maps a Y -vector into an N_{tot} -vector and takes history sizes into account. To obtain dimensions compatible with other vectors and matrices, we define $\boldsymbol{\omega} = \mathbf{R}_0 \boldsymbol{\omega}^Y$ and $\bar{\boldsymbol{\omega}} = \mathbf{R}_1 \boldsymbol{\omega}^Y$.

In conclusion, the two constraints of Definition 1 on the weights $\boldsymbol{\omega}^Y = (\omega_y)_y$ are:

$$\mathbf{L}_1 \boldsymbol{\omega}^Y = \mathbf{L}_2 \boldsymbol{\omega}^Y, \quad (120)$$

with $\mathbf{L}_1 = \tilde{\mathbf{L}}_1 \mathbf{R}_1$ and $\mathbf{L}_2 = \tilde{\mathbf{L}}_2 \mathbf{R}_1$.

E Changes in the fiscal system

To isolate how the fiscal system affects recovered weights, we swap fiscal systems between the two countries: France adopts the US fiscal system and vice versa. Table 8 summarizes the new fiscal system for each country.

	United States	France
τ_k	0.35	0.36
τ_c	0.18	0.05
κ	0.98	0.65
τ	0.23	0.16
B/Y	0.21	0.91

Table 8: New fiscal system for the United States and France in the current experiment.

We use our estimation strategy to compute the new G-SWF weights with the updated fiscal system of Table 8. We report in Figure 7 the differences implied by the new fiscal system compared to the benchmark for some key variables. These variables are G-SWF weights, utility level (including labor disutility), labor income, and change in capital income. The results are averaged for each productivity level.

Panel (a) shows the US economy under the French fiscal system. Replacing the fiscal system raises recovered weights for low-productivity agents and lowers them for high-productivity agents. The mechanism is visible in period utility: low-productivity agents gain from the more redistributive system, while high-productivity agents face lower utility because labor and capital income taxation become more progressive.

Panel (b) shows the reverse experiment for France under the US fiscal system. Low-productivity agents lose from the less redistributive fiscal system, which lowers their recovered G-SWF weights. High-productivity agents gain, raising the upper tail of the recovered profile.

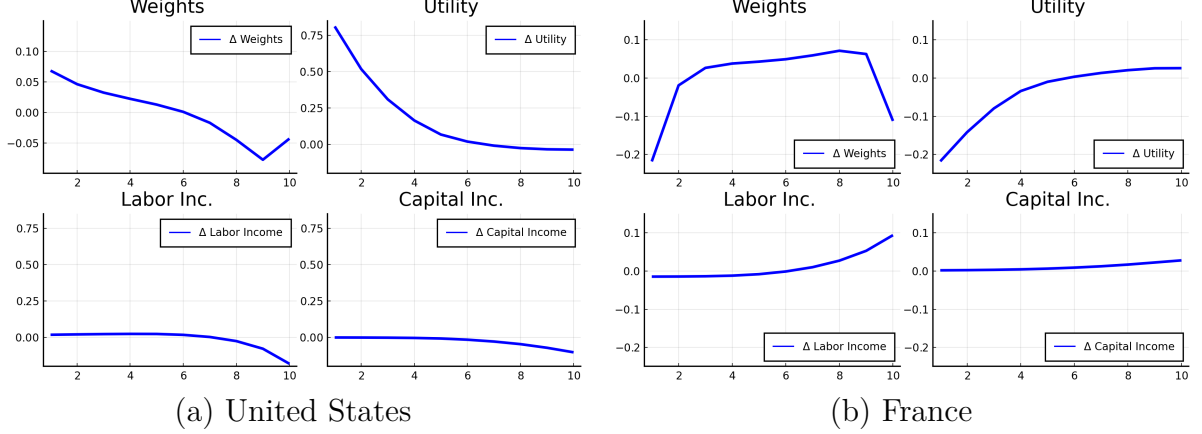


Figure 7: Differences in recovered weights and selected outcomes after swapping fiscal systems.

F Robustness checks for G-SWF weights

This appendix reports three checks around the recovered G-SWF weights and related objects. First, we relax the log-quadratic parametric form used in the baseline (Section F.1). Second, we allow weights to depend on truncated histories rather than only on current productivity (Section F.2). Finally, we report implied marginal weights that combine recovered G-SWF weights with average marginal utilities in Section F.3.

F.1 Non-parametric weights

In the baseline quantitative exercise, we estimated parametric weights by imposing a functional relationship between the weights and productivity to obtain an exact identification (see Definition 1). We describe here the non-parametric recovery of G-SWF weights. Without the log-quadratic restriction, the Ramsey constraints and normalization leave the weight vector underidentified. We therefore select, among admissible weight vectors, the profile with minimum dispersion around the utilitarian benchmark.

As explained in Section 3.3 (see equation (120)), the Ramsey FOCs impose two constraints: $\sum_{y \in \mathcal{Y}} L_{k,y} \omega_y = 0$, where $L_{k,y} \in \mathbb{R}$ ($k = 1, 2$ and $y \in \mathcal{Y}$). The variance-

minimizing weights are characterized by the vector $(\hat{\omega}_y)_y$, solving the following program:

$$(\hat{\omega}_y)_y = \underset{(\omega_y)_y}{\operatorname{argmin}} \sum_{y \in \mathcal{Y}} \pi_y (\omega_y - 1)^2, \quad (121)$$

$$\text{s.t. } 0 = \sum_{y \in \mathcal{Y}} L_{k,y} \omega_y \text{ for all } k = 1, 2, \quad (122)$$

$$1 = \sum_{y \in \mathcal{Y}} \pi_y \omega_y. \quad (123)$$

F.2 Weights per truncated history

We relax current-productivity sufficiency by allowing recovered weights to depend on the whole truncated history. The number of unknown weights then rises from Y to Y^N , so the Ramsey restrictions leave the weights highly underidentified. We use the same minimum-variance selection criterion as in the non-parametric exercise of Section F.1. Formally, the weights $(\hat{\omega}_{y^N})_{y^N}$ are determined as follows:

$$(\hat{\omega}_{y^N})_{y^N} = \underset{(\omega_{y^N})_{y^N}}{\operatorname{argmin}} \sum_{y^N \in \mathcal{Y}^N} S_{y^N} (\omega_{y^N} - 1)^2, \quad (124)$$

$$\text{s.t. } 0 = \sum_{y^N \in \mathcal{Y}^N} \tilde{L}_{k,y^N} \omega_{y^N} \text{ for all } k = 1, 2, \quad (125)$$

$$1 = \sum_{y^N \in \mathcal{Y}^N} S_{y^N} \omega_{y^N}, \quad (126)$$

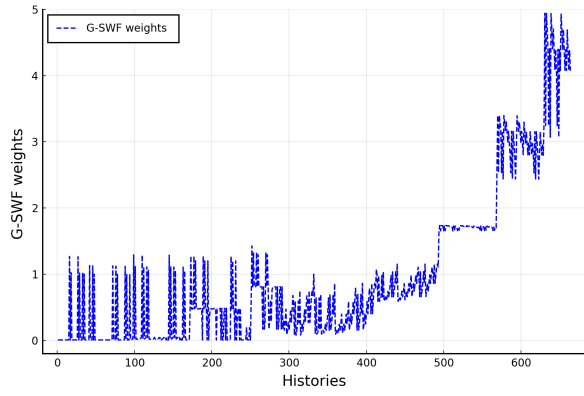
where $\tilde{L}_1$ and $\tilde{L}_2$ are defined in (118) and (119).

Figure 8 plots the recovered truncated-history weights for the United States in panel (a) and France in panel (b). We restrict attention to histories with a positive mass. To compare these high-dimensional weights with the baseline G-SWF, we average them within each current-productivity state using the stationary mass of each truncated history. Figure 9 compares the resulting average history weights with the non-parametric current-productivity weights. The two profiles are close in both countries, indicating that the recovered weight information is preserved when the weight space is expanded to truncated histories.

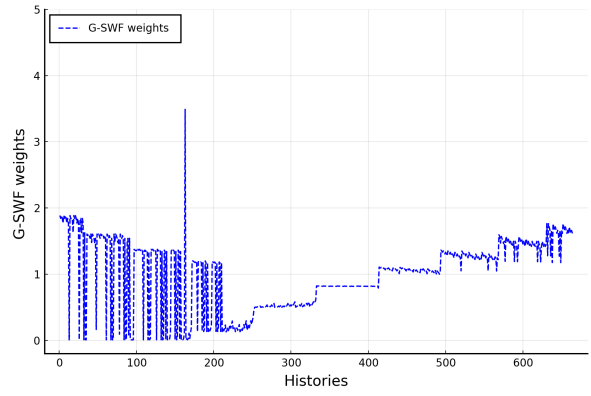
F.3 Implied marginal weights

This appendix reports implied social marginal welfare weights, denoted $(\lambda_y)_y$, which combine recovered G-SWF weights with average marginal utilities of consumption: $\lambda_y = \omega_y \bar{u}'(c_y)$. These objects differ from the recovered planner weights: they measure the social value of a marginal resource transfer at each productivity level, whereas the G-SWF weights measure the planner's valuation of utility units. Figure 10 reports λ_y by income quintile, normalized to average one, for both countries.

Two observations emerge. First, despite the marked differences in G-SWF weights

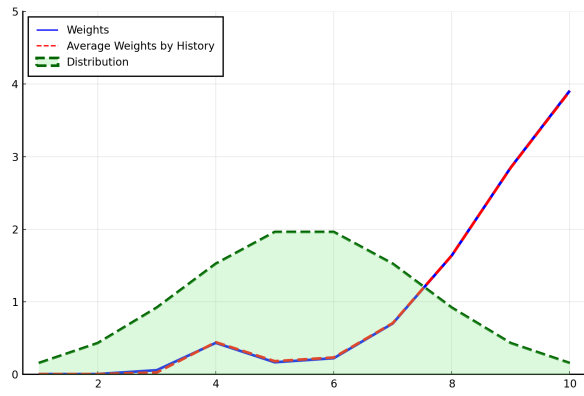


(a) United States

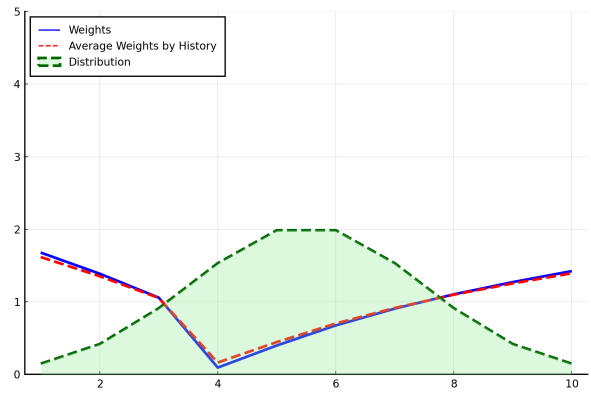


(b) France

Figure 8: History weights for the United States and France. Histories are arranged in ascending order of current productivity.



(a) United States



(b) France

Figure 9: Comparison between non-parametric G-SWF weights and average history weights for the United States and France.

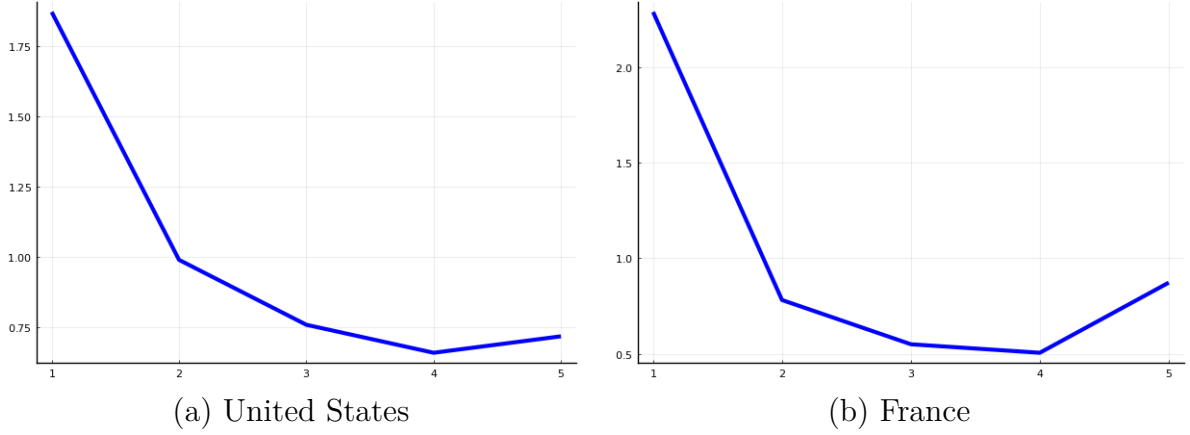


Figure 10: Implied marginal weights $\lambda_y = \omega_y \bar{u}'(c_y)$ by labor-income quintile in the United States and France.

between the two countries (Figure 1), the implied marginal weights λ_y display broadly decreasing patterns in both the United States and France. Because average marginal utility declines sharply with productivity, implied marginal weights can be decreasing even when recovered G-SWF weights are not. This distinction is why the main text reports G-SWF weights separately from implied marginal weights.

Second, the US profile is close to that estimated by Hendren (2020), who recovers analogous objects from administrative data on transfers: decreasing across the income distribution with a slight uptick at the top quintile. Our first-quintile value (around 1.8) exceeds Hendren’s (1.2), reflecting that the stylized fiscal system used here does not capture all transfers that compress marginal utilities at the bottom of the distribution in administrative data. The broad agreement nonetheless supports the internal consistency of the calibrated model with microeconomic evidence.

G Robustness to the preference parameter σ

This appendix details the robustness exercise that varies the inverse intertemporal elasticity of substitution. For each value of $\sigma \in \{1, 2, 2.5\}$, we recalibrate the income process and fiscal instruments to match the same steady-state targets as in the baseline, then recover G-SWF weights using the same inverse-optimal procedure. The goal is to test whether the country-level weight profiles survive changes in utility curvature once the model is recalibrated.

The income-process parameters (ϱ, σ^y) are searched jointly with the fiscal instruments (τ, κ, t_c) over a four-dimensional grid. For each candidate combination, κ is determined by bisection to pin G/Y at its target, and the objective minimizes the squared distance to the remaining wealth and Gini targets. The winning parameters are re-solved at full precision ($n_a = 100$, tolerance 10^{-8}), after which the truncation projections and recovery

of G-SWF weights proceed identically to the respective country baseline.

Table 9 lists the parameters held fixed across the sigma robustness exercises.

Table 9: Parameters held fixed across the σ robustness exercises.

Parameter	Symbol	United States	France
Capital share	α	0.36	0.36
Quarterly depreciation	δ	0.025	0.025
Frisch elasticity	φ	0.5	0.5
Capital tax rate	t_k	0.36	0.35
Annual capital-output ratio	K/Y	2.7	3.1
Average hours target	$\bar{l}$	1/3.5	1/5.8

Tables 10 and 11 report the calibrated parameters and achieved moments for the United States and France, respectively.

Table 10: Calibrated parameters and steady-state moments, United States.

σ	Parameters					Moments				
	ϱ	σ^y	τ	κ	t_c	B/Y	Gini	G/Y	C/Y	I/Y
Target	—	—	—	—	—	0.63	0.78	0.15	0.58	0.27
1.8 (baseline)	0.962	0.196	0.16	0.850	0.05	0.63	0.78	0.15	0.58	0.27
1	0.962	0.230	0.06	0.925	0.13	0.58	0.83	0.15	0.58	0.27
2	0.962	0.196	0.22	0.869	0.09	0.55	0.77	0.16	0.57	0.27
2.5	0.965	0.196	0.24	0.831	0.05	0.58	0.77	0.15	0.58	0.27

Table 11: Calibrated parameters and steady-state moments, France.

σ	Parameters					Moments				
	ϱ	σ^y	τ	κ	t_c	B/Y	Gini	G/Y	C/Y	I/Y
Target	—	—	—	—	—	0.61	0.68	0.25	0.44	0.31
1.8 (baseline)	0.951	0.127	0.23	0.728	0.18	0.61	0.68	0.25	0.44	0.31
1	0.930	0.127	0.10	0.892	0.42	-0.05	0.66	0.25	0.44	0.31
2	0.951	0.127	0.29	0.759	0.23	0.45	0.68	0.25	0.44	0.31
2.5	0.957	0.127	0.32	0.673	0.10	0.43	0.69	0.25	0.44	0.31

For the United States, the flow ratios (G/Y , C/Y , I/Y) remain close to their targets across the grid. The wealth Gini matches the target for the baseline and higher-curvature cases, but rises to 0.83 under log utility. Public debt remains below the 0.63 target in the alternative calibrations (0.58, 0.55, and 0.58), consistent with the recalibration using debt as a residual margin. For France, all flow targets and the Gini are closely matched for $\sigma \in \{2, 2.5\}$. Under log utility ($\sigma = 1$), the lower precautionary-savings motive makes

it structurally difficult to sustain a positive B/Y within a realistic fiscal range, and the calibration yields $B/Y = -0.05$; all other moments remain close to target.

Figures 11 and 12 compare the G-SWF weights for the alternative values of σ directly with the baseline $\sigma = 1.8$ calibration (dashed black line).

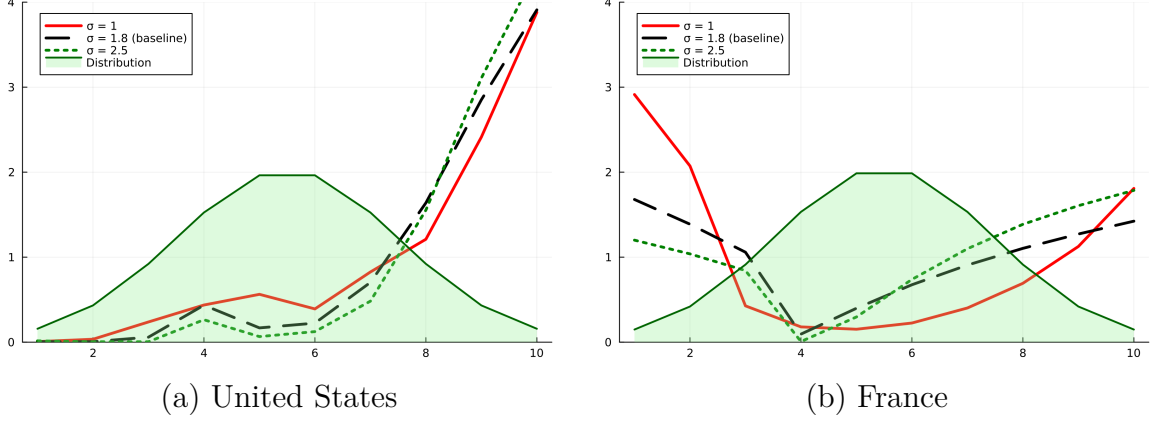


Figure 11: Non-parametric G-SWF weights for $\sigma \in \{1, 2.5\}$ and the baseline $\sigma = 1.8$ (dashed black line), United States and France.

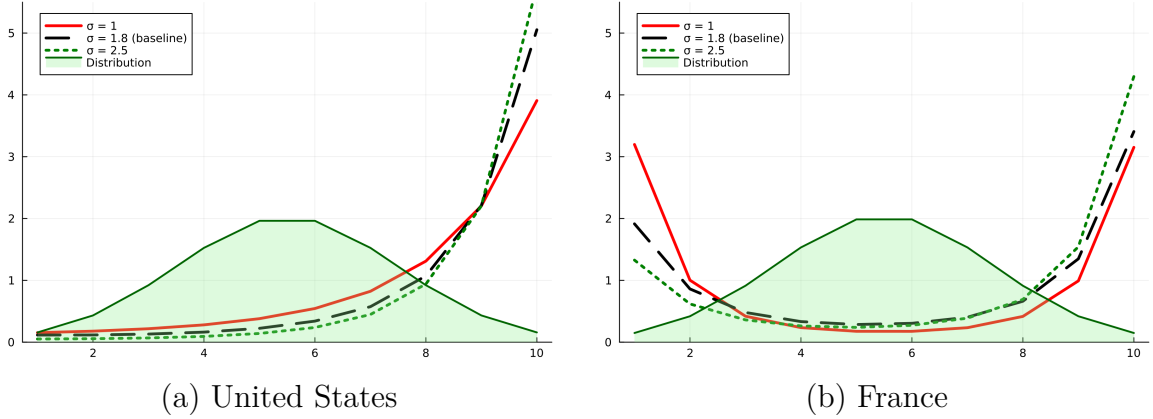


Figure 12: Parametric G-SWF weights for $\sigma \in \{1, 2.5\}$ and the baseline $\sigma = 1.8$ (dashed black line), United States and France.

The alternative- σ profiles preserve the same country-level shapes in both the parametric and non-parametric recoveries. In the United States, the profiles remain increasing in productivity, with level differences concentrated near the upper tail. In France, the profiles remain non-monotone and U-shaped, although the height of the lower tail varies with curvature. These comparisons support the main-text conclusion that the qualitative country profiles are not driven by the baseline calibration, as discussed in Section 6.2.

H Robustness to the Frisch elasticity φ

This appendix repeats the robustness exercise for the Frisch elasticity. We vary the Frisch elasticity $\varphi \in \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50\}$, hold the income process and inverse IES at their country-specific baseline values, recalibrate the fiscal instruments to match the same steady-state targets, and adjust the labor disutility coefficient χ to keep average hours aligned with the country target.

Table 9 lists the parameters held fixed in the Frisch-elasticity exercise, except φ , which is varied in this exercise. The curvature σ and the country-specific income-process parameters remain at their baseline values.

Tables 12 and 13 report the calibrated parameters and achieved moments for the United States and France, respectively.

Table 12: Calibrated parameters and steady-state moments, United States.

φ	Parameters			Moments				
	τ	κ	t_c	B/Y	Gini	G/Y	C/Y	I/Y
Target	—	—	—	0.63	0.78	0.15	0.58	0.27
0.25	0.20	0.888	0.11	0.60	0.77	0.15	0.58	0.27
0.50 (baseline)	0.16	0.850	0.05	0.55	0.78	0.15	0.58	0.27
0.75	0.14	0.831	0.03	0.57	0.79	0.15	0.58	0.27
1.00	0.12	0.831	0.03	0.68	0.79	0.15	0.58	0.27
1.25	0.12	0.831	0.03	0.55	0.80	0.15	0.58	0.27
1.50	0.12	0.850	0.05	0.53	0.81	0.15	0.58	0.27

Table 13: Calibrated parameters and steady-state moments, France.

φ	Parameters			Moments				
	τ	κ	t_c	B/Y	Gini	G/Y	C/Y	I/Y
Target	—	—	—	0.61	0.68	0.25	0.44	0.31
0.25	0.24	0.681	0.16	0.50	0.67	0.25	0.44	0.31
0.50 (baseline)	0.23	0.697	0.13	0.45	0.68	0.25	0.44	0.31
0.75	0.23	0.728	0.16	0.47	0.69	0.25	0.44	0.31
1.00	0.23	0.744	0.16	0.48	0.70	0.25	0.44	0.31
1.25	0.23	0.744	0.16	0.44	0.71	0.25	0.44	0.31
1.50	0.23	0.759	0.16	0.51	0.71	0.25	0.44	0.31

The recalibrations keep the flow targets on target in both countries. The remaining movement is concentrated in debt and the wealth Gini. In the United States, the Gini rises moderately as labor becomes more elastic and public debt B/Y fluctuates around its target. In France, public debt B/Y remains below target across the grid, as in the

baseline calibration. These calibration residuals are small relative to the stability of the recovered G-SWF profiles reported below.

Figures 13 and 14 display the G-SWF weights for all six values of φ , with panel (a) for the United States and panel (b) for France. In each figure, the shaded green area shows the stationary income distribution (scaled for visibility).

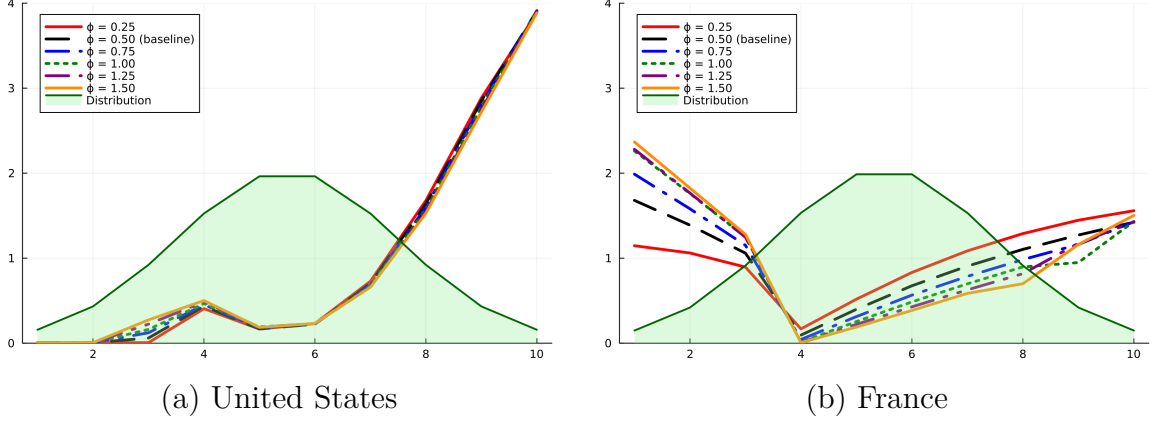


Figure 13: Non-parametric G-SWF weights ω_y for $\varphi \in \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50\}$ and income distribution (green shaded area). The x-axis indexes the 10 productivity levels in ascending order.

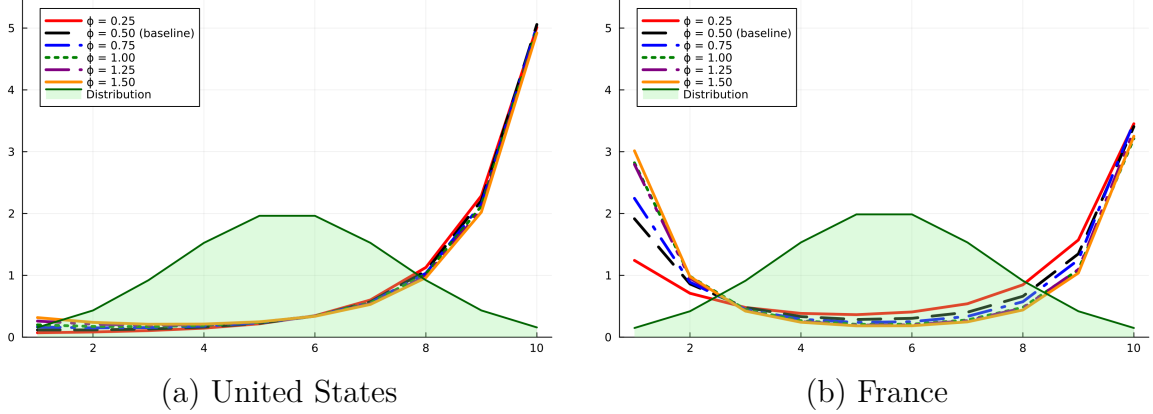


Figure 14: Parametric G-SWF weights ω_y for $\varphi \in \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50\}$ and income distribution (green shaded area).

The qualitative profiles are stable across all values of φ . In the United States, the profiles remain tilted toward high-productivity states and are nearly indistinguishable across the grid. In France, the non-monotone U-shape is preserved, although tail heights move somewhat with the elasticity.

Across the parametric and non-parametric recoveries of Figures 14 and 13, the same country-level qualitative patterns remain visible. The Frisch exercise therefore reinforces the IES one: within the preference specifications we consider, the recovered social valuations

are shaped mainly by the tax-and-transfer architecture rather than by the preference specification.